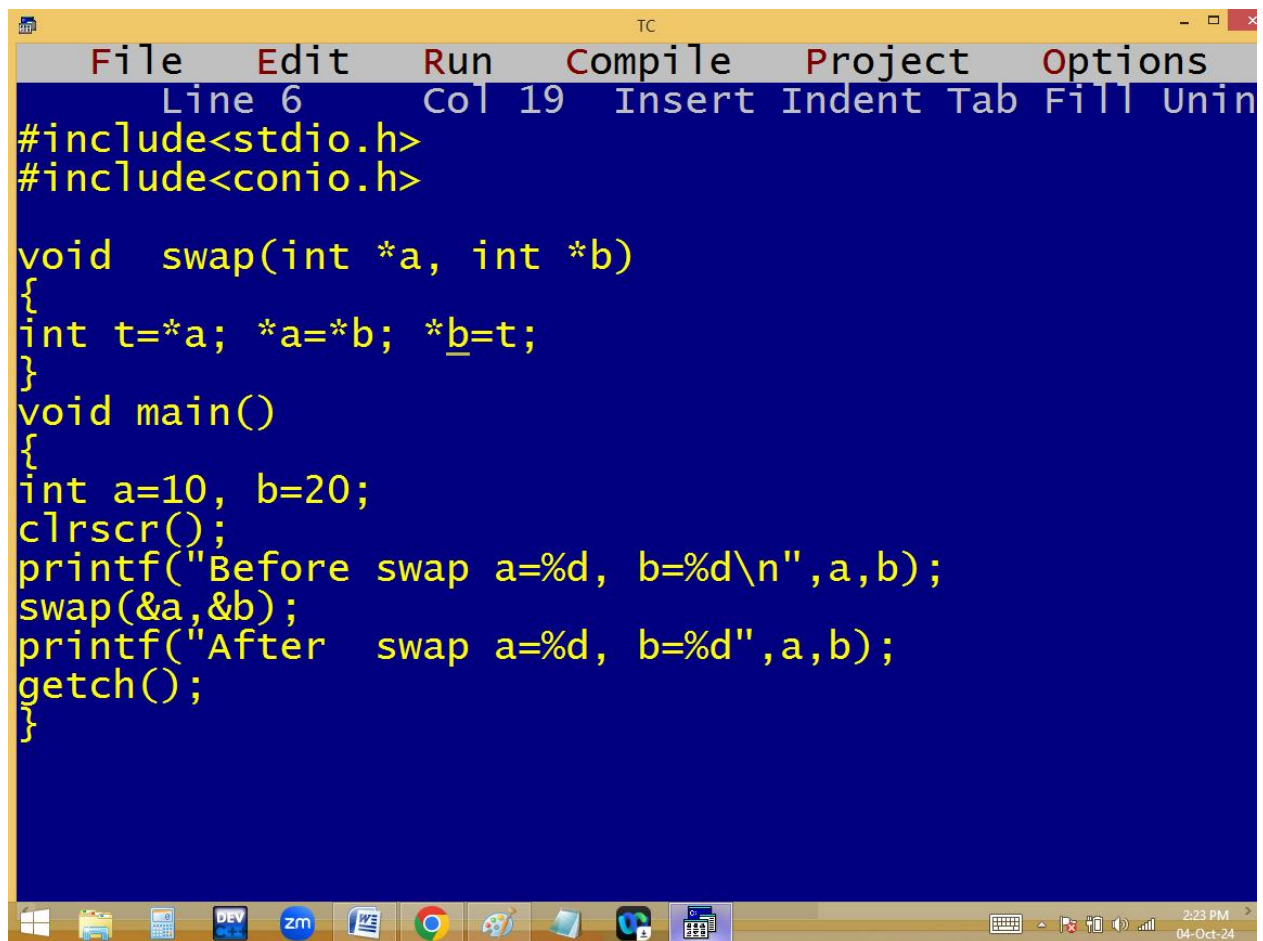




The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, and Options. Under the "Run" menu, there is a submenu with the following options: Line 6, Col 19, Insert, Indent, Tab, Fill, and Unin. The main editing area has a dark blue background with yellow text. The code is as follows:

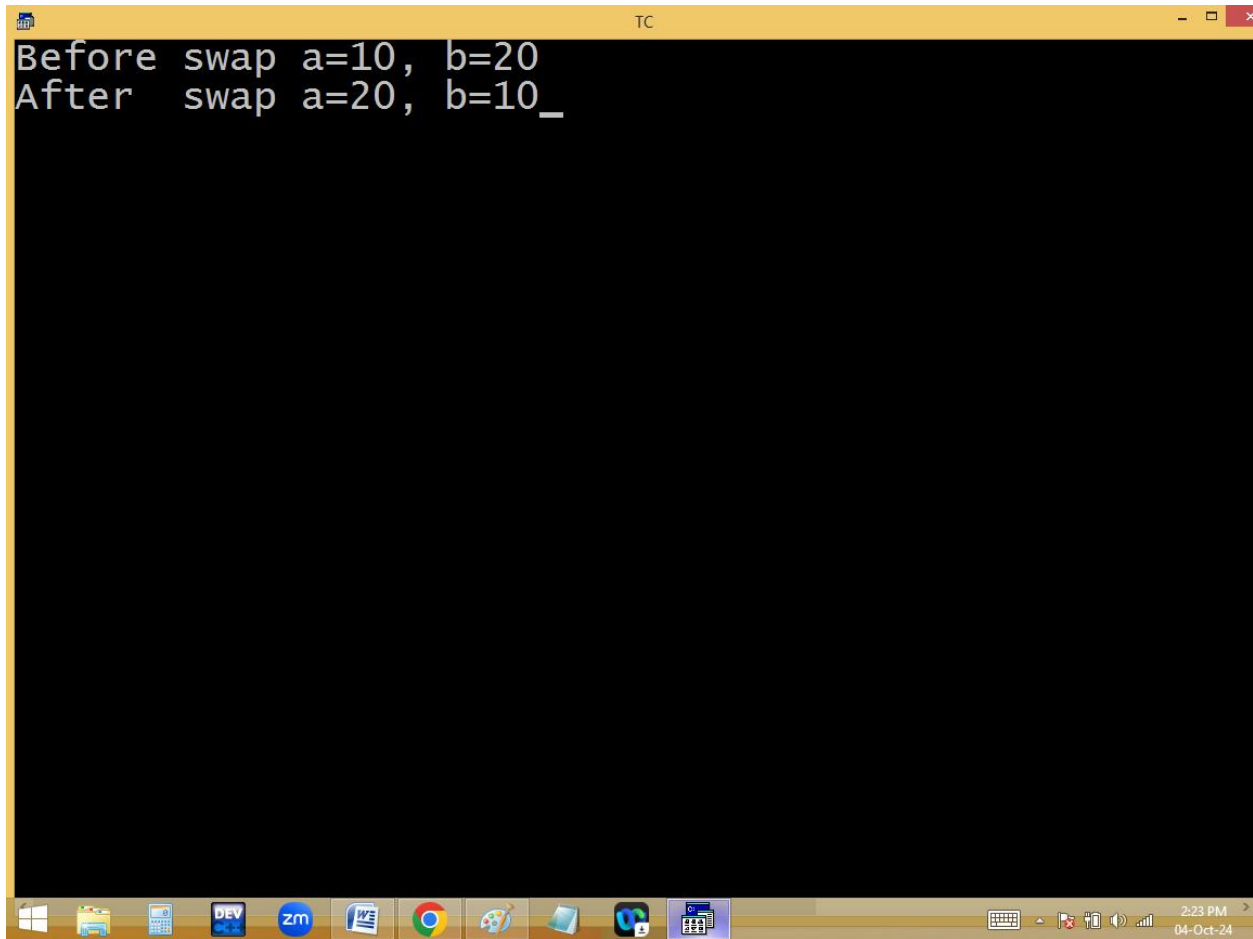
```
#include<stdio.h>
#include<conio.h>

void swap(int *a, int *b)
{
    int t=*a; *a=*b; *b=t;
}

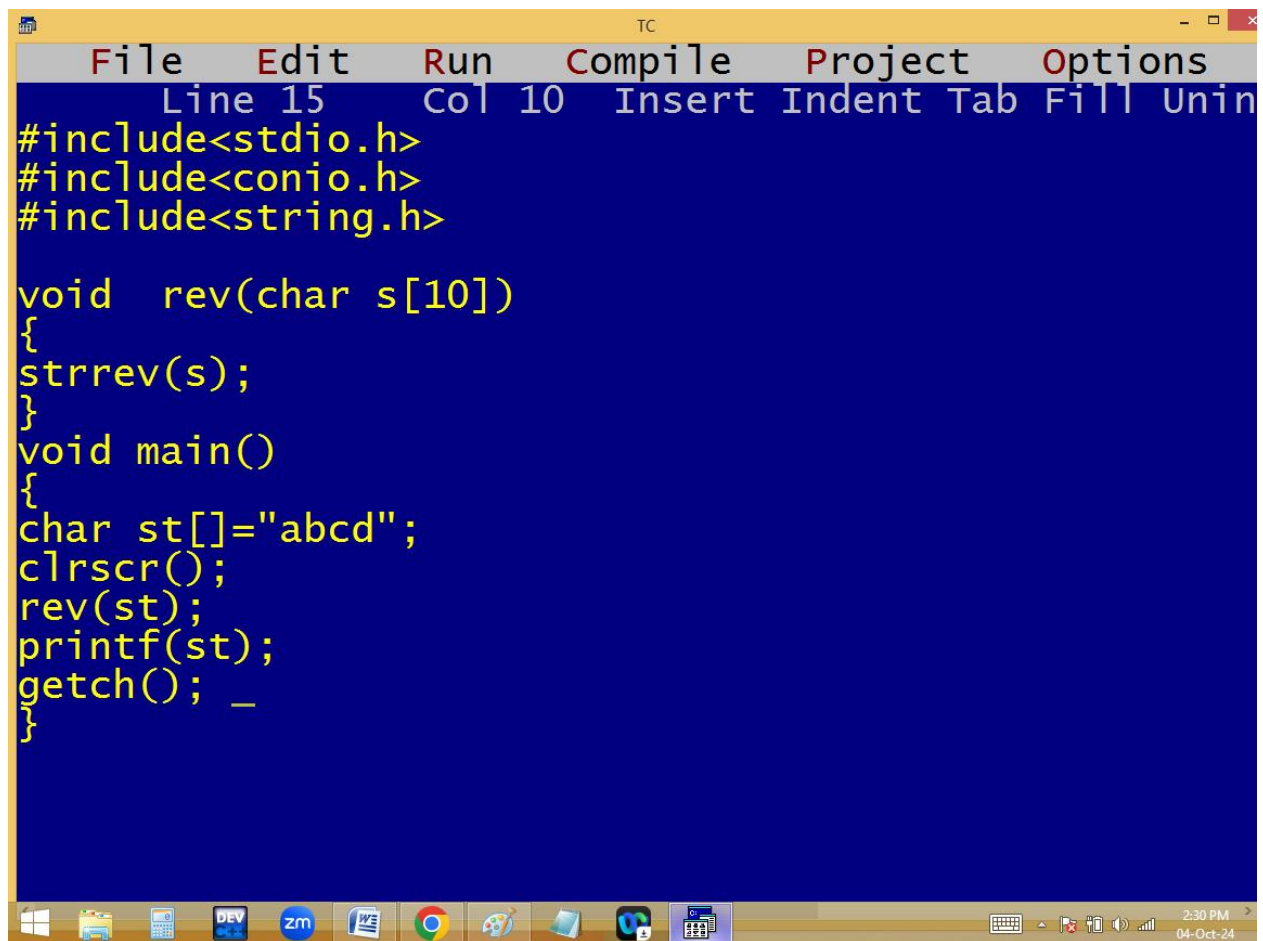
void main()
{
    int a=10, b=20;
    clrscr();
    printf("Before swap a=%d, b=%d\n",a,b);
    swap(&a,&b);
    printf("After swap a=%d, b=%d",a,b);
    getch();
}
```

At the bottom of the window is a Windows taskbar. It contains several icons: the Windows Start button, a folder icon, a calculator icon, a "DEV" icon, a "zm" icon, a "WZ" icon, a Google Chrome icon, a paint icon, a file explorer icon, a VS Code icon, and a calendar icon. On the right side of the taskbar, there is a system tray showing the time as "2:23 PM" and the date as "04-Oct-24".

```
TC
Before swap a=10, b=20
After swap a=20, b=10_
```

A screenshot of a Windows desktop environment. At the top, a yellow title bar for a window titled 'TC' is visible. The main area of the window is black, displaying white text: 'Before swap a=10, b=20' on the first line and 'After swap a=20, b=10_' on the second line. The Windows taskbar is at the bottom, showing various application icons including Windows Explorer, Calculator, DEV C++, Zoom, Word, Chrome, Paint, File Explorer, and a calendar. The system tray on the right shows the time as 2:23 PM and the date as 04-Oct-24.

Passing array or string to function:



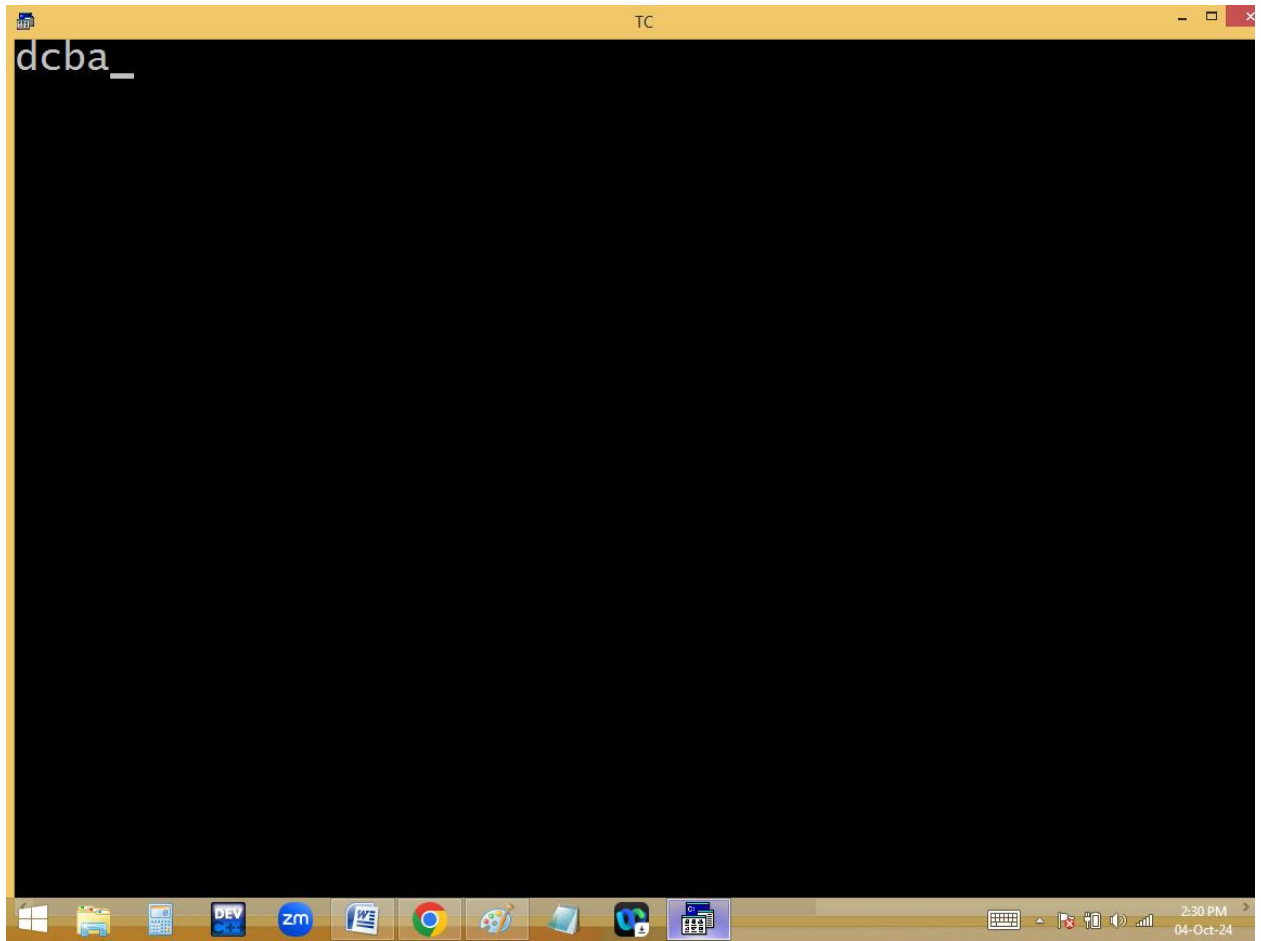
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, and Options. Under the "Run" menu, there is a submenu with the following options: Line 15, Col 10, Insert, Indent, Tab, Fill, and Unin. The main editing area has a dark blue background with yellow text. The code is as follows:

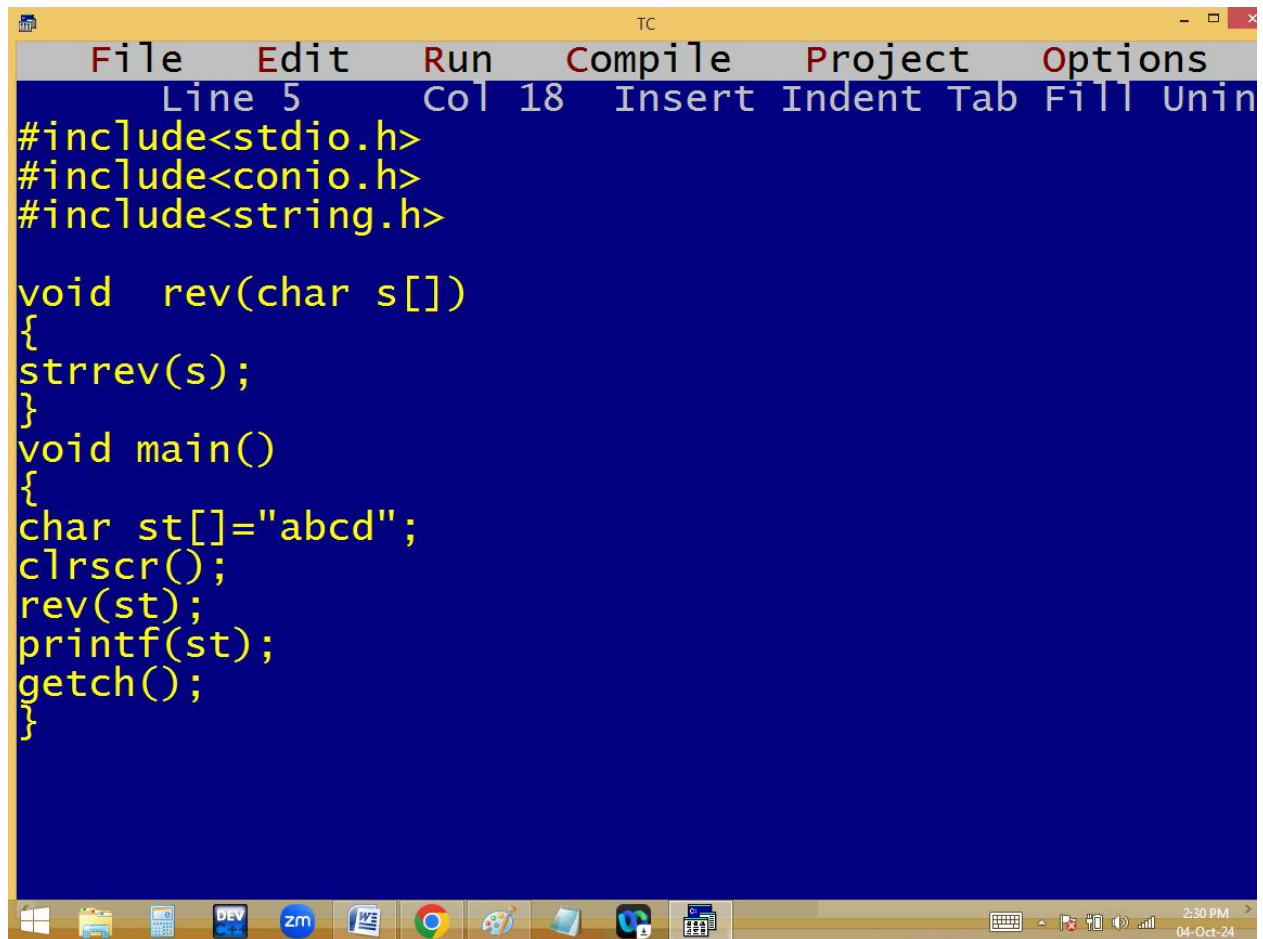
```
#include<stdio.h>
#include<conio.h>
#include<string.h>

void rev(char s[10])
{
    strrev(s);
}

void main()
{
    char st[]="abcd";
    clrscr();
    rev(st);
    printf(st);
    getch(); _
}
```

At the bottom of the window is a Windows taskbar. It contains several icons: the Windows Start button, a folder icon, a calculator icon, a "DEV" icon, a "zm" icon, a "WZ" icon, a Google Chrome icon, a paint icon, a file explorer icon, a VS Code icon, and a calendar icon. On the right side of the taskbar, there are system tray icons for keyboard, volume, and network, along with the system clock showing "2:30 PM" and the date "04-Oct-24".





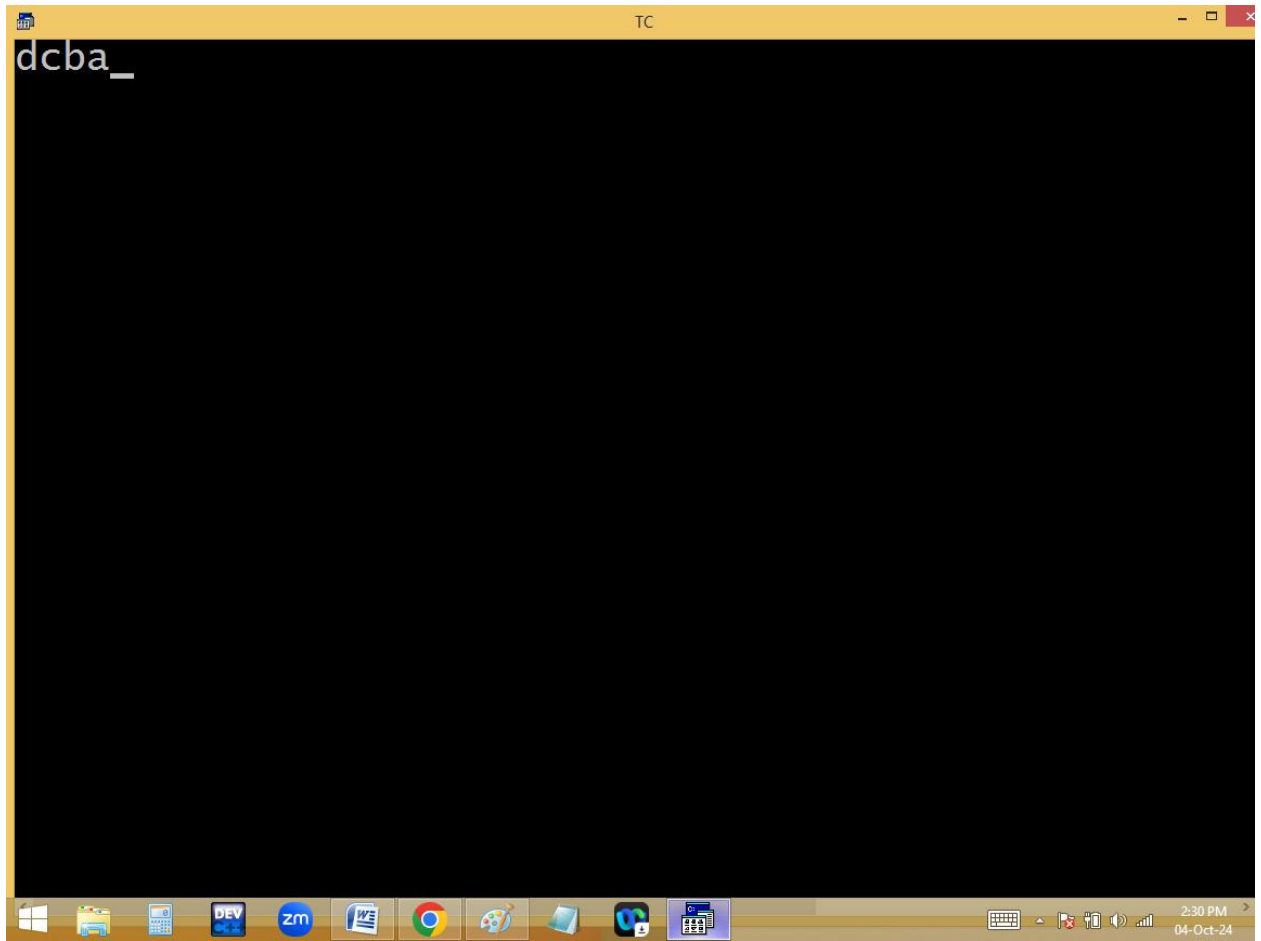
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, and Options. Under the "Run" menu, there is a submenu with the following options: Line 5, Col 18, Insert, Indent, Tab, Fill, and Unin. The main editing area has a dark blue background with yellow text. The code is as follows:

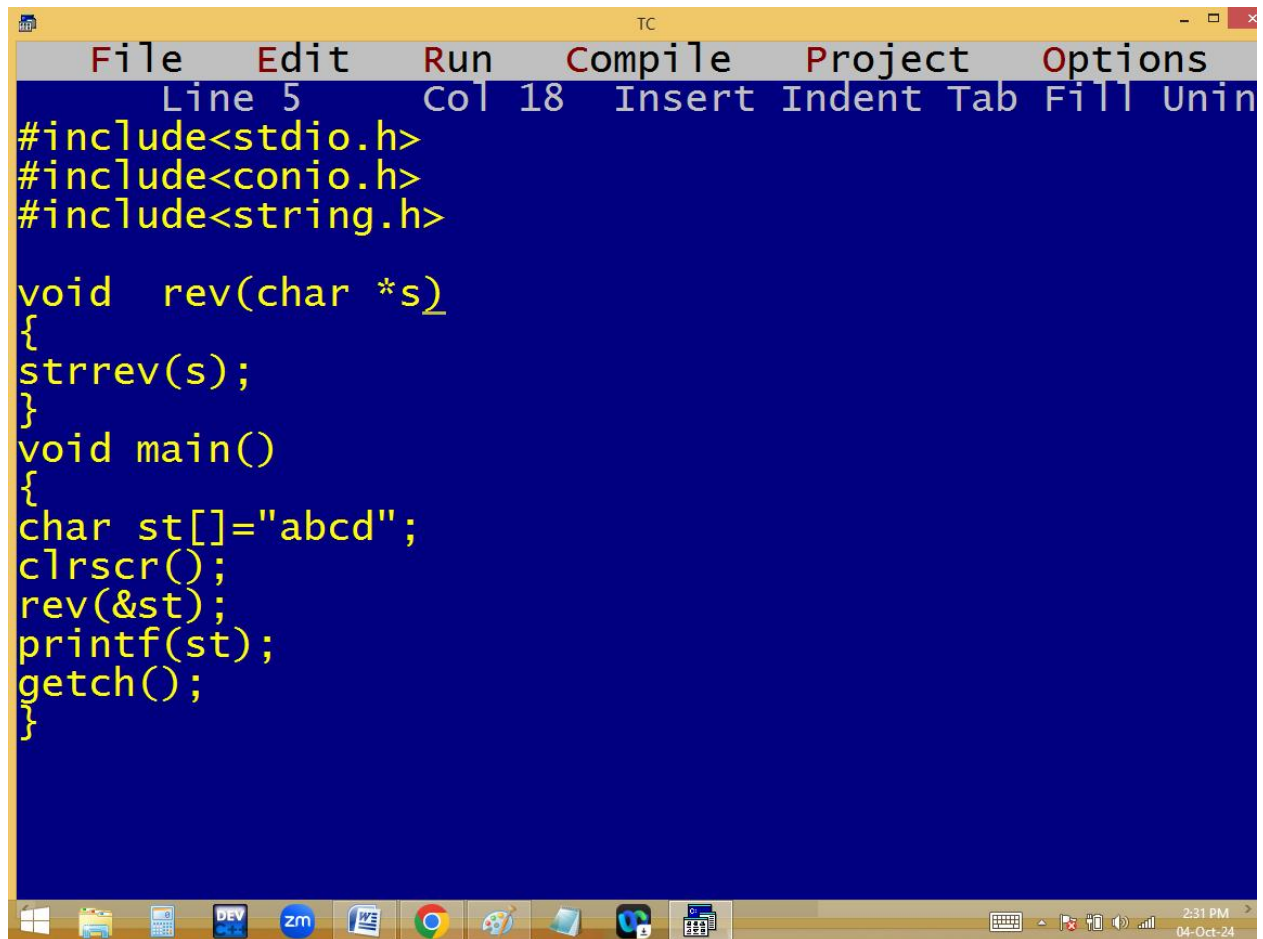
```
#include<stdio.h>
#include<conio.h>
#include<string.h>

void rev(char s[])
{
    strrev(s);
}

void main()
{
    char st[]="abcd";
    clrscr();
    rev(st);
    printf(st);
    getch();
}
```

At the bottom of the window is a Windows taskbar. It contains several icons: the Windows Start button, a folder icon, a calculator icon, a Dev C++ icon, a Zoom icon, a Word icon, a Chrome icon, a Paint icon, a File Explorer icon, and a Task Manager icon. On the right side of the taskbar, there is a system tray showing the time as 2:30 PM and the date as 04-Oct-24.





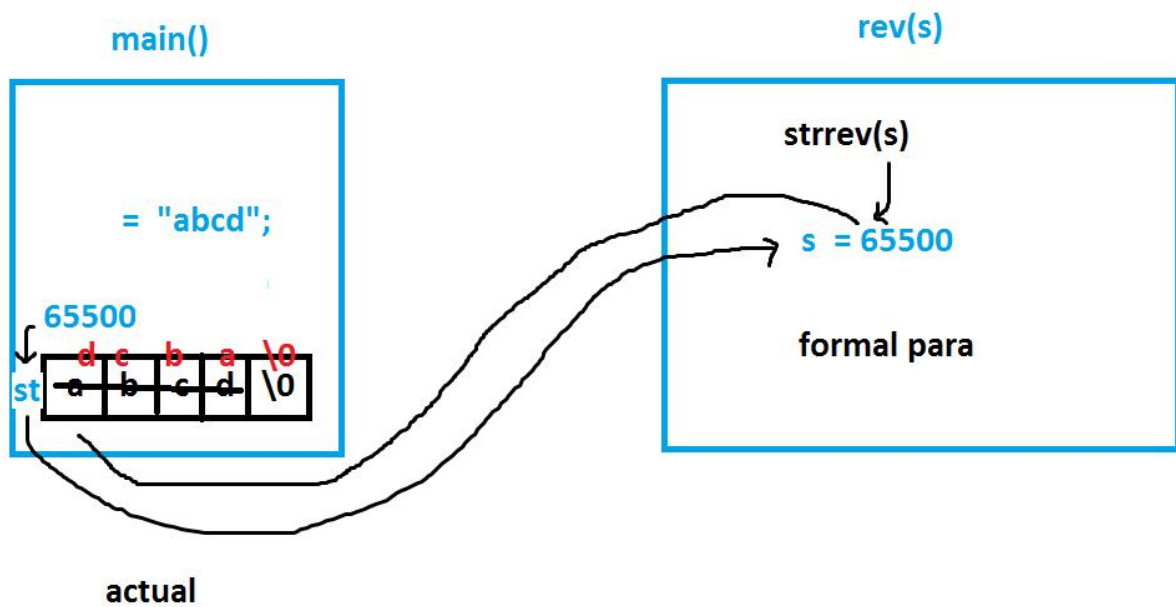
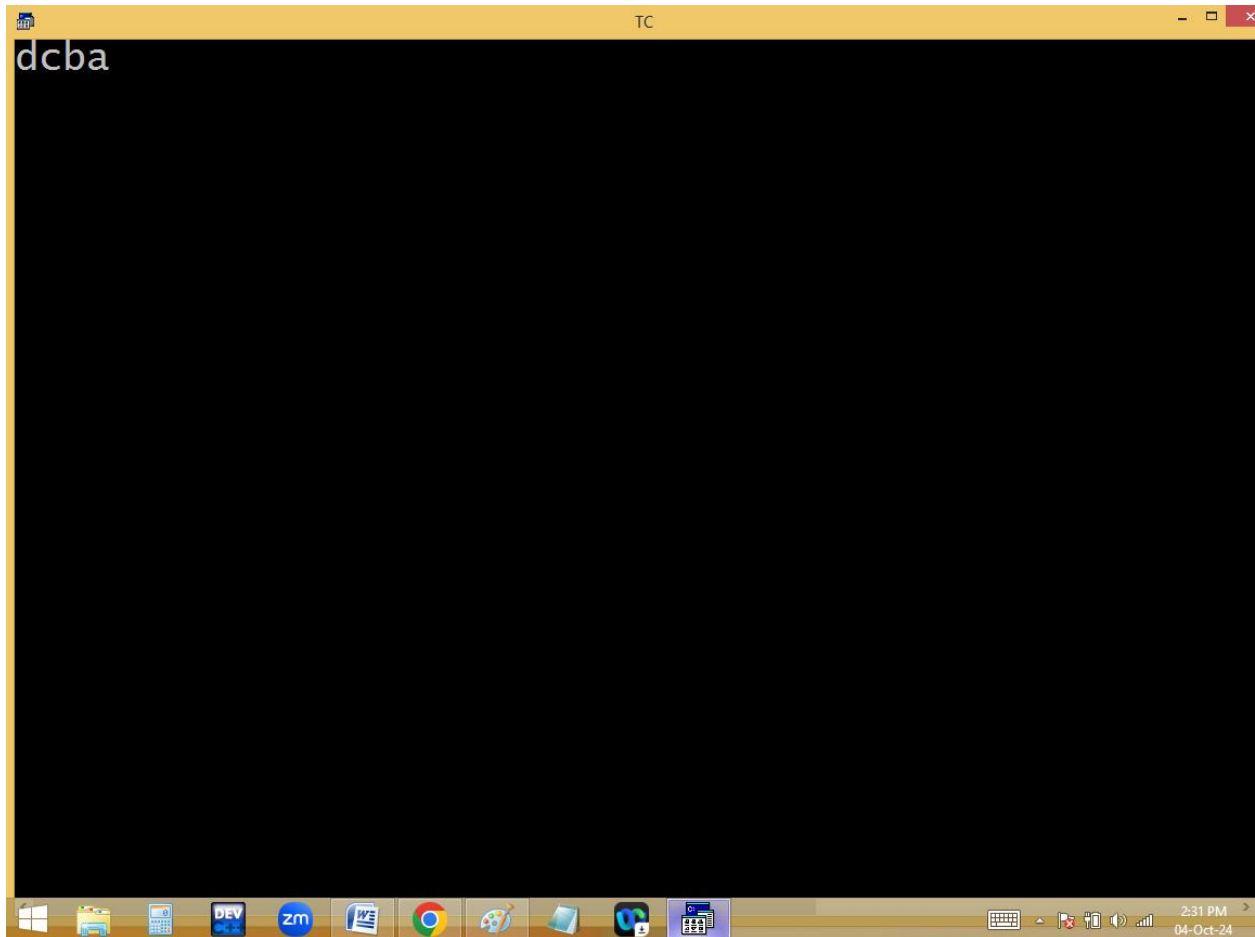
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". Below it is a menu bar with the following options: File, Edit, Run, Compile, Project, and Options. A status bar at the top of the editor area displays "Line 5", "Col 18", and "Insert Indent Tab Fill Unin". The main editor area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
#include<string.h>

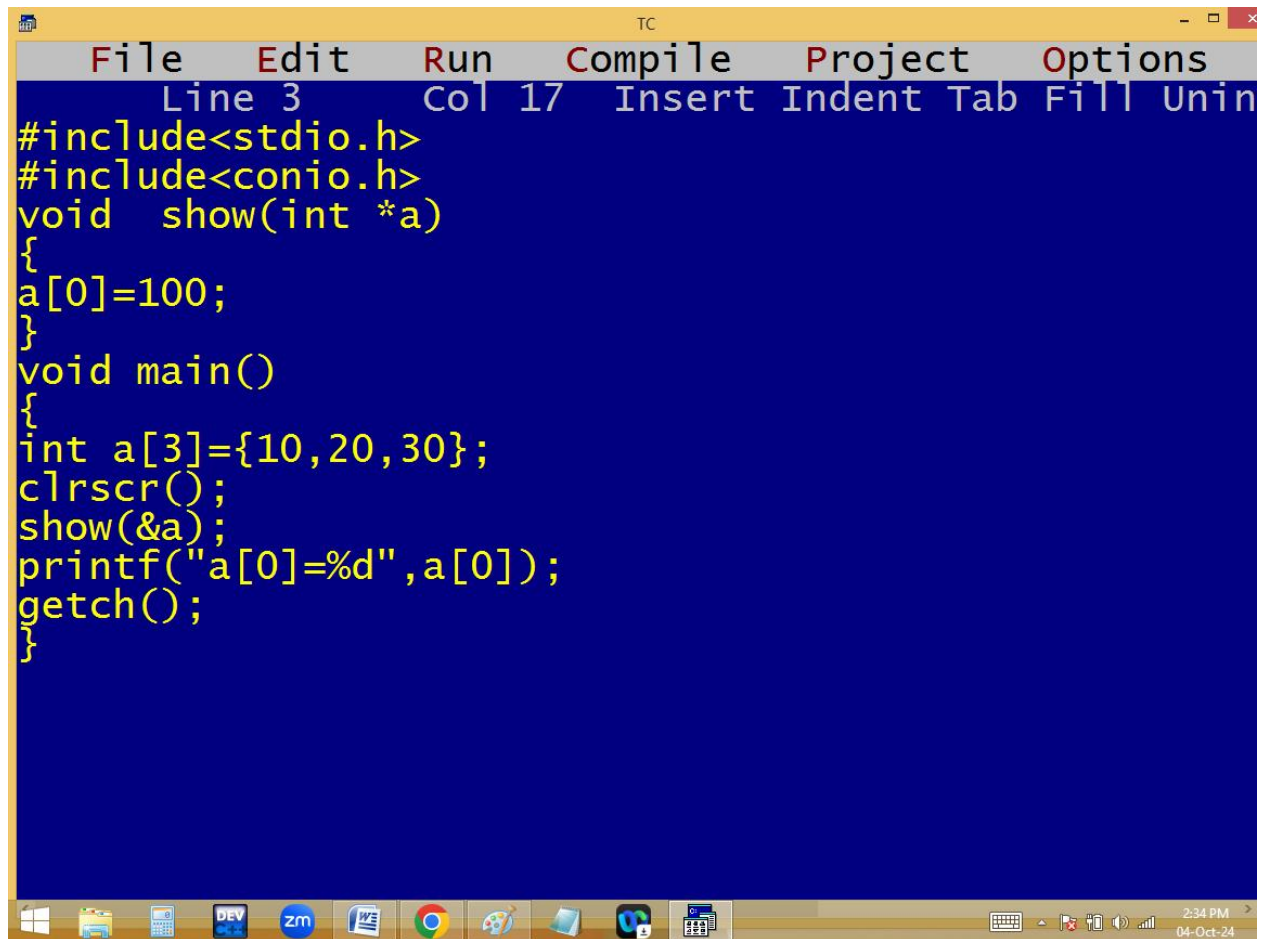
void rev(char *s)
{
    strrev(s);
}

void main()
{
    char st[]="abcd";
    clrscr();
    rev(&st);
    printf(st);
    getch();
}
```

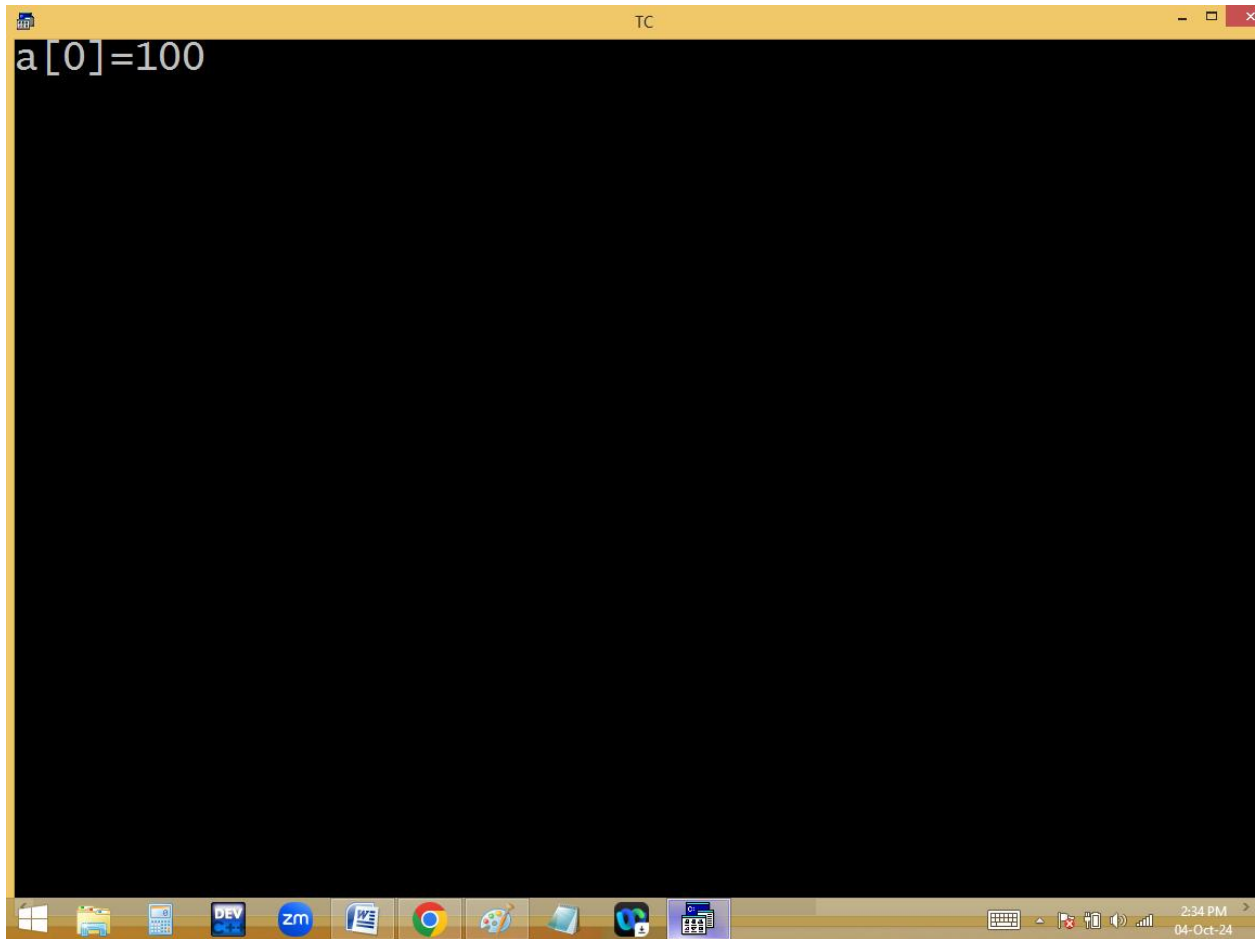
At the bottom of the window is a Windows taskbar. It contains several icons: the Start button, File Explorer, Calculator, DEV C++, Zoom (zm), Word, Google Chrome, Paint, a folder icon, a game icon, and a calendar icon. On the right side of the taskbar, there are system tray icons for keyboard, network, volume, and battery, along with the date and time: "2:31 PM" and "04-Oct-24".

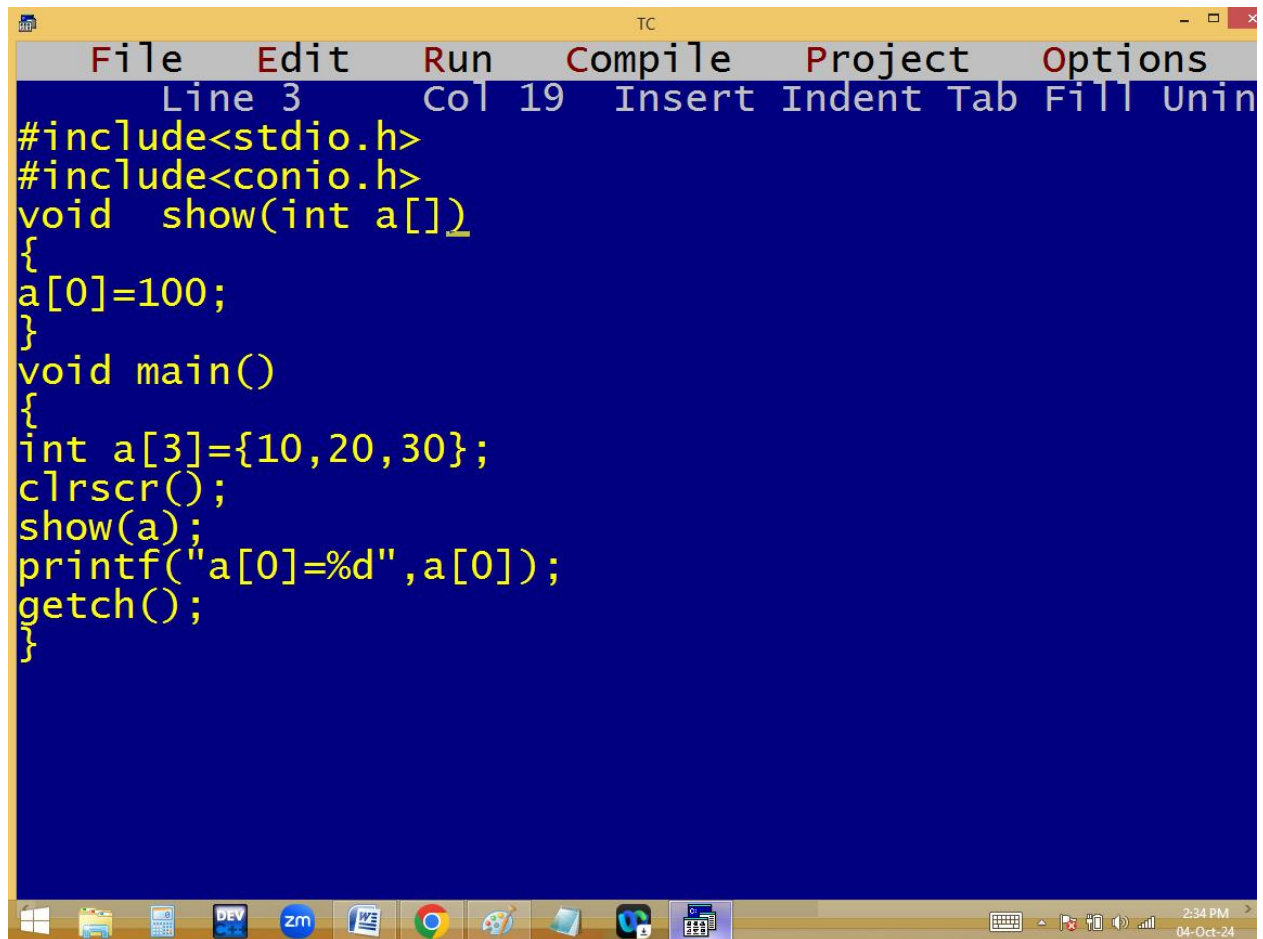


Passing array to function:



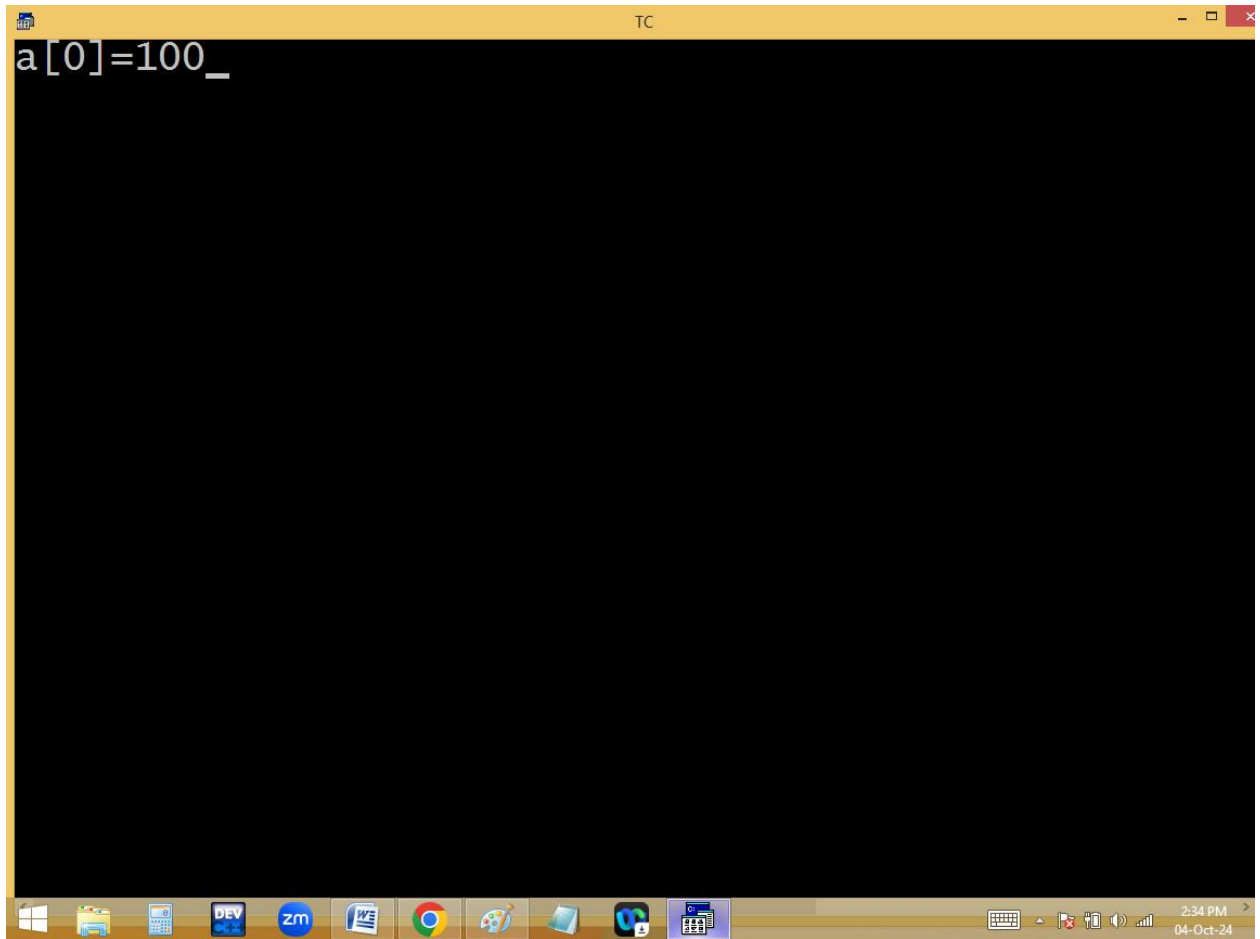
```
TC
File Edit Run Compile Project Options
Line 3 Col 17 Insert Indent Tab Fill Unin
#include<stdio.h>
#include<conio.h>
void show(int *a)
{
a[0]=100;
}
void main()
{
int a[3]={10,20,30};
clrscr();
show(&a);
printf("a[0]=%d",a[0]);
getch();
}
```

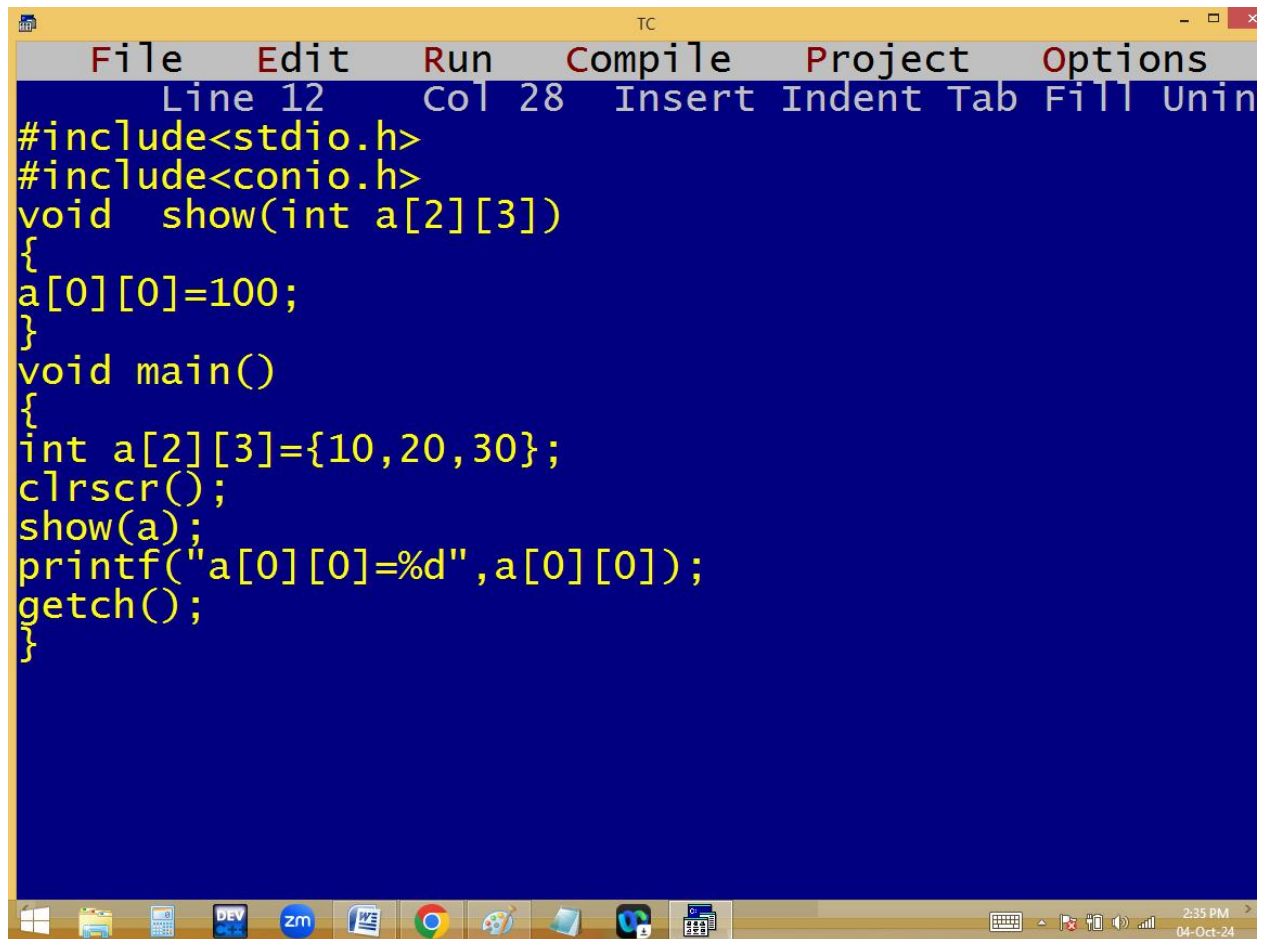




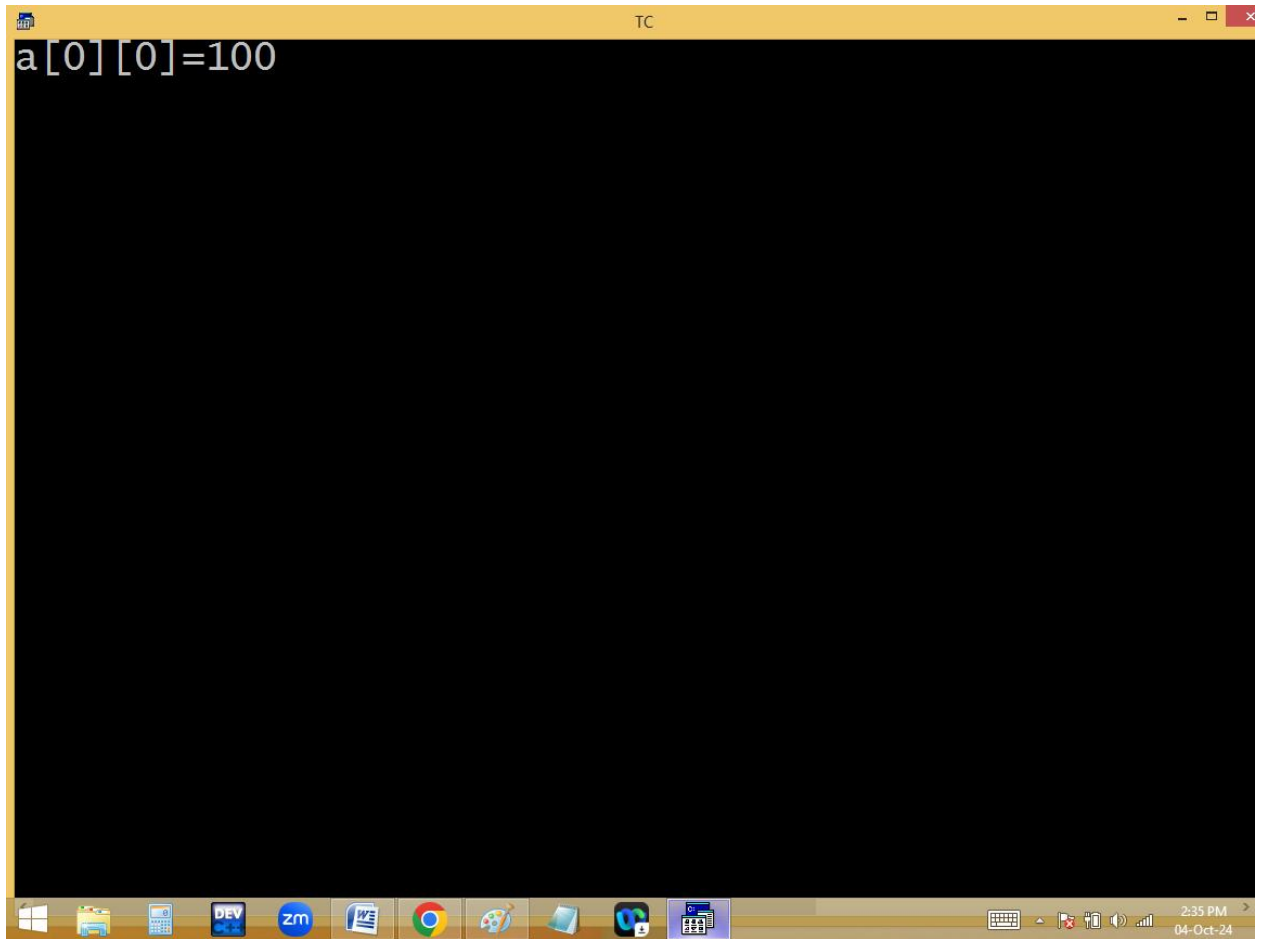
```
TC
File Edit Run Compile Project Options
Line 3 Col 19 Insert Indent Tab Fill Unin
#include<stdio.h>
#include<conio.h>
void show(int a[])
{
a[0]=100;
}
void main()
{
int a[3]={10,20,30};
clrscr();
show(a);
printf("a[0]=%d",a[0]);
getch();
}
```

2:34 PM
04-Oct-24





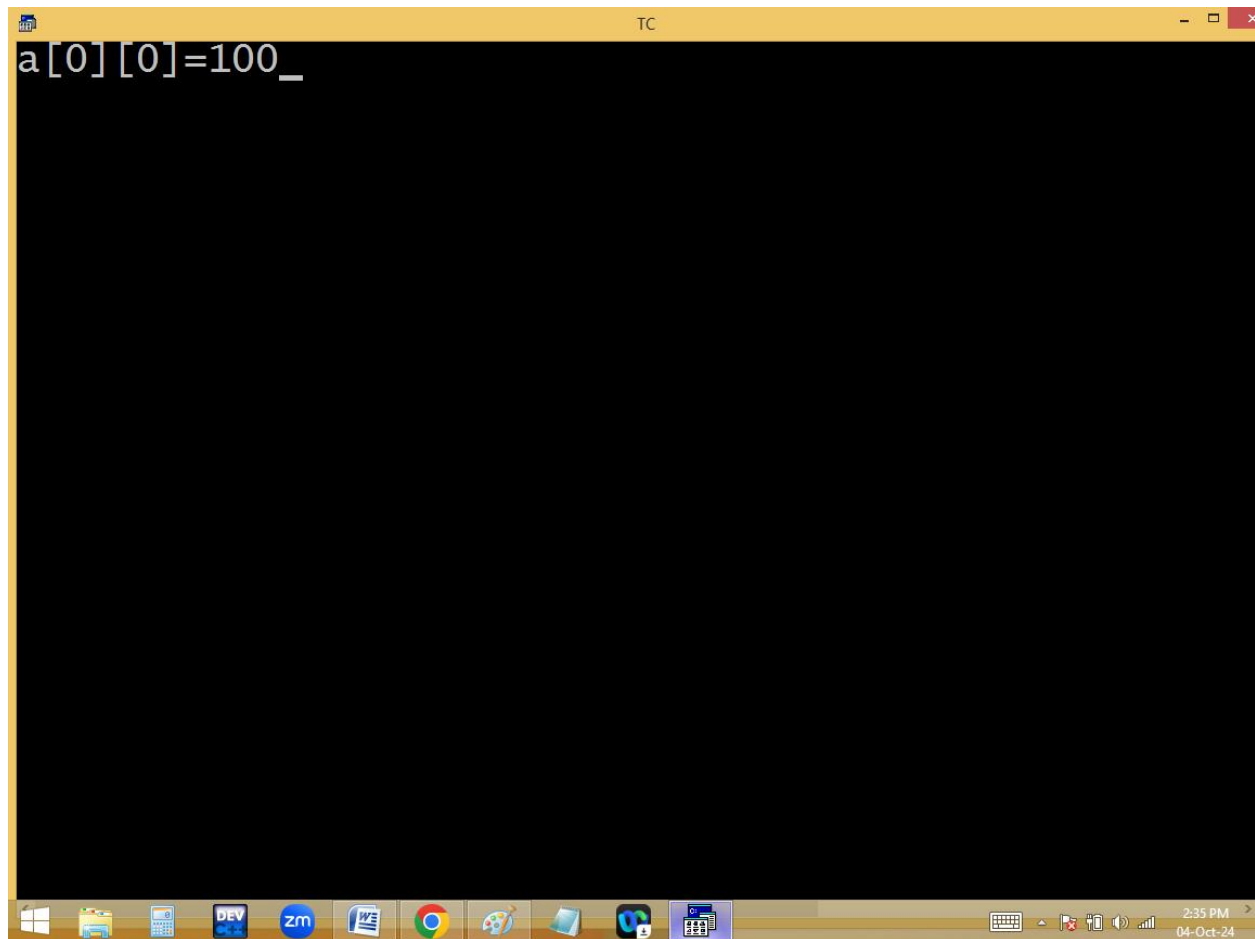
```
TC
File Edit Run Compile Project Options
Line 12 Col 28 Insert Indent Tab Fill Unin
#include<stdio.h>
#include<conio.h>
void show(int a[2][3])
{
a[0][0]=100;
}
void main()
{
int a[2][3]={10,20,30};
clrscr();
show(a);
printf("a[0][0]=%d",a[0][0]);
getch();
}
```



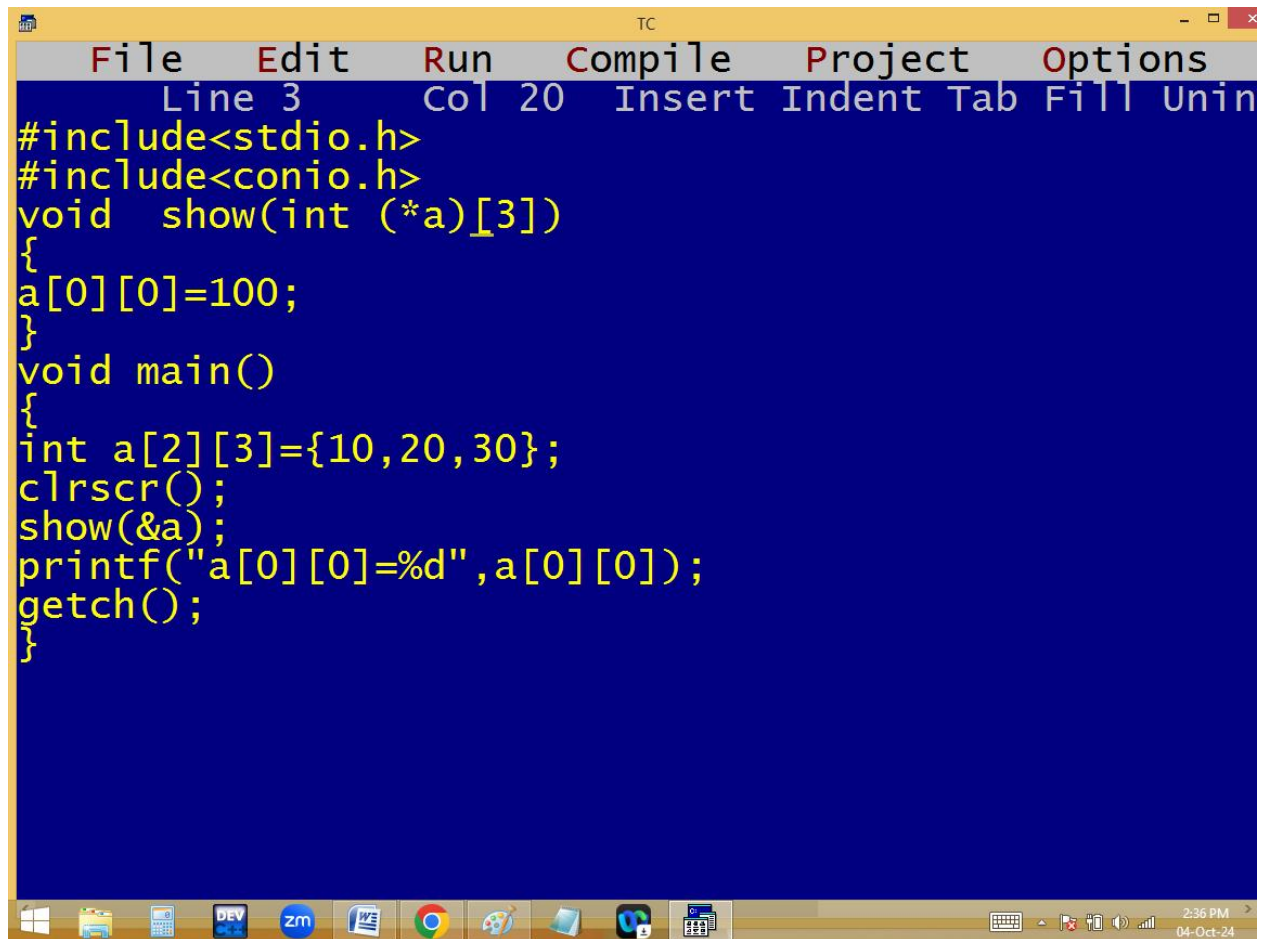
```
TC
File Edit Run Compile Project Options
Line 3 Col 21 Insert Indent Tab Fill Unin
#include<stdio.h>
#include<conio.h>
void show(int a[][3])
{
a[0][0]=100;
}
void main()
{
int a[2][3]={10,20,30};
clrscr();
show(a);
printf("a[0][0]=%d",a[0][0]);
getch();
}
```

2:35 PM
04-Oct-24

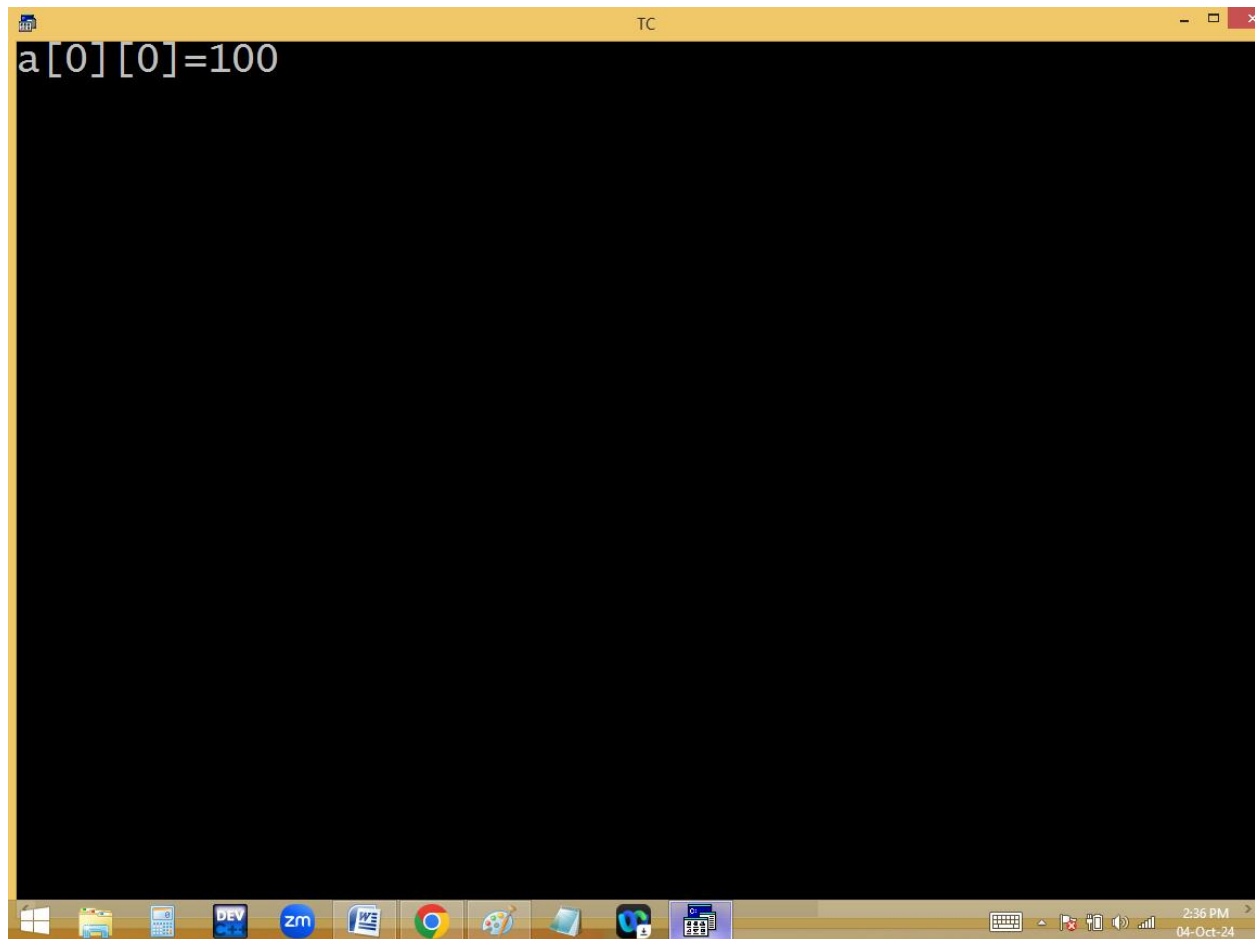

```
TC
a[0][0]=100_
```



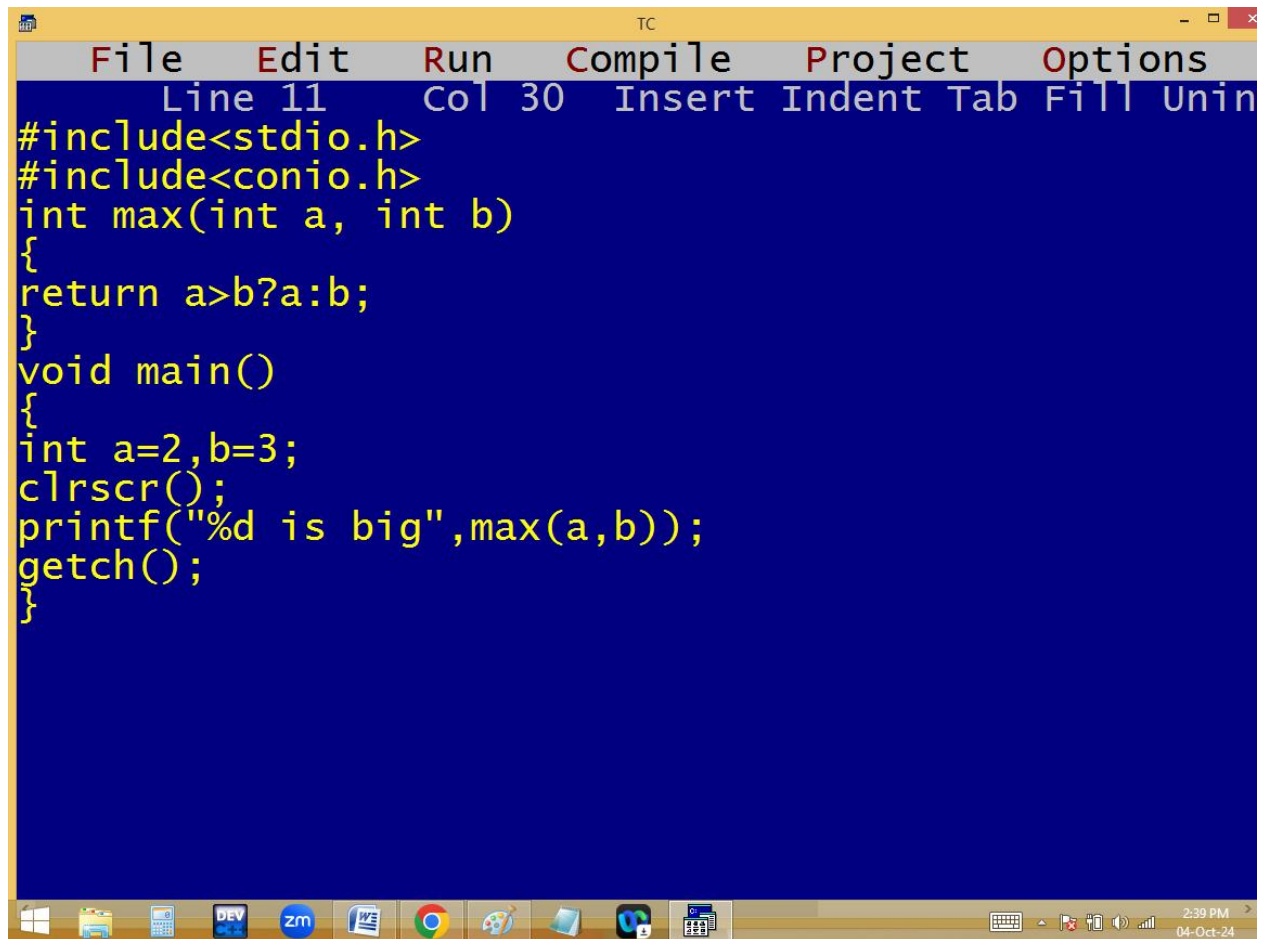
Passing a 2 dimensional array to function:



```
TC
File Edit Run Compile Project Options
Line 3 Col 20 Insert Indent Tab Fill Unin
#include<stdio.h>
#include<conio.h>
void show(int (*a)[3])
{
a[0][0]=100;
}
void main()
{
int a[2][3]={10,20,30};
clrscr();
show(&a);
printf("a[0][0]=%d",a[0][0]);
getch();
}
```



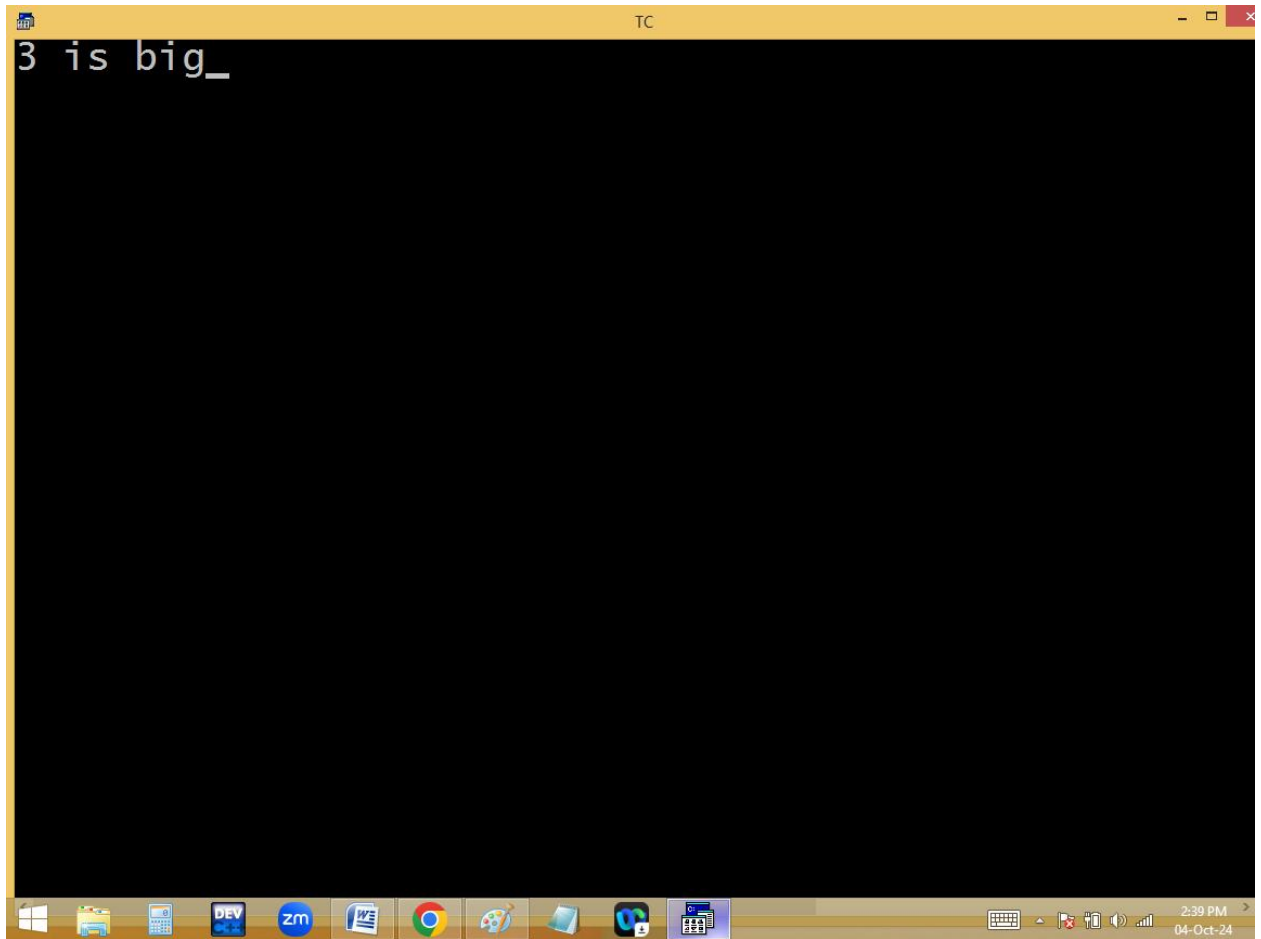
Function returning address [pointer]:

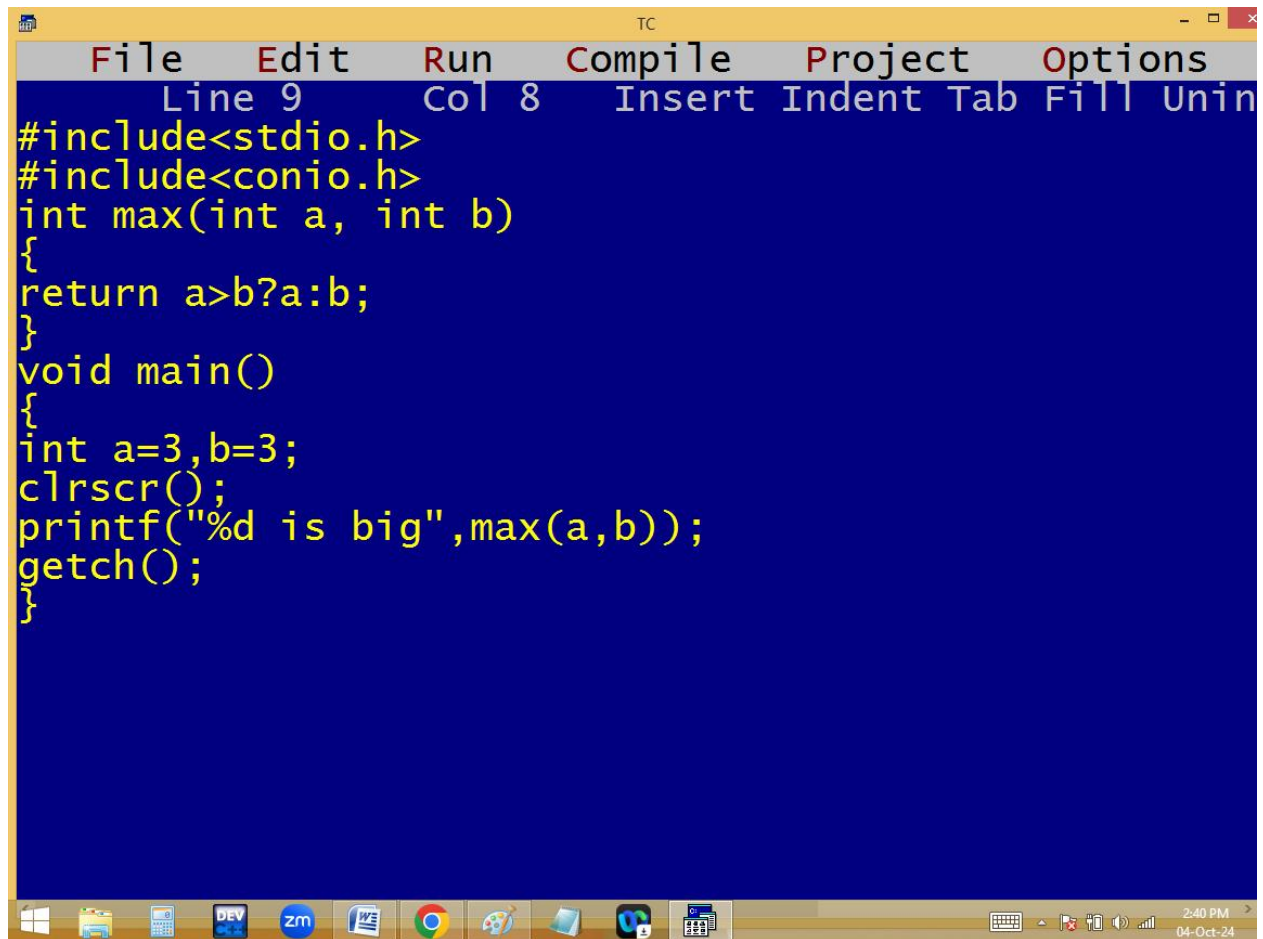


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, and Options. Under the "Run" menu, there is a submenu with the following options: Line 11, Col 30, Insert, Indent, Tab, Fill, and Unin. The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
int max(int a, int b)
{
    return a>b?a:b;
}
void main()
{
    int a=2,b=3;
    clrscr();
    printf("%d is big",max(a,b));
    getch();
}
```

At the bottom of the window is a Windows taskbar. It contains several icons: the Windows Start button, a folder icon, a calculator icon, a "DEV" icon, a "zm" icon, a "WZ" icon, the Google Chrome icon, a paint icon, a file explorer icon, a VS Code icon, and a calendar icon. On the right side of the taskbar, there is a system tray showing the time as "2:39 PM" and the date as "04-Oct-24".

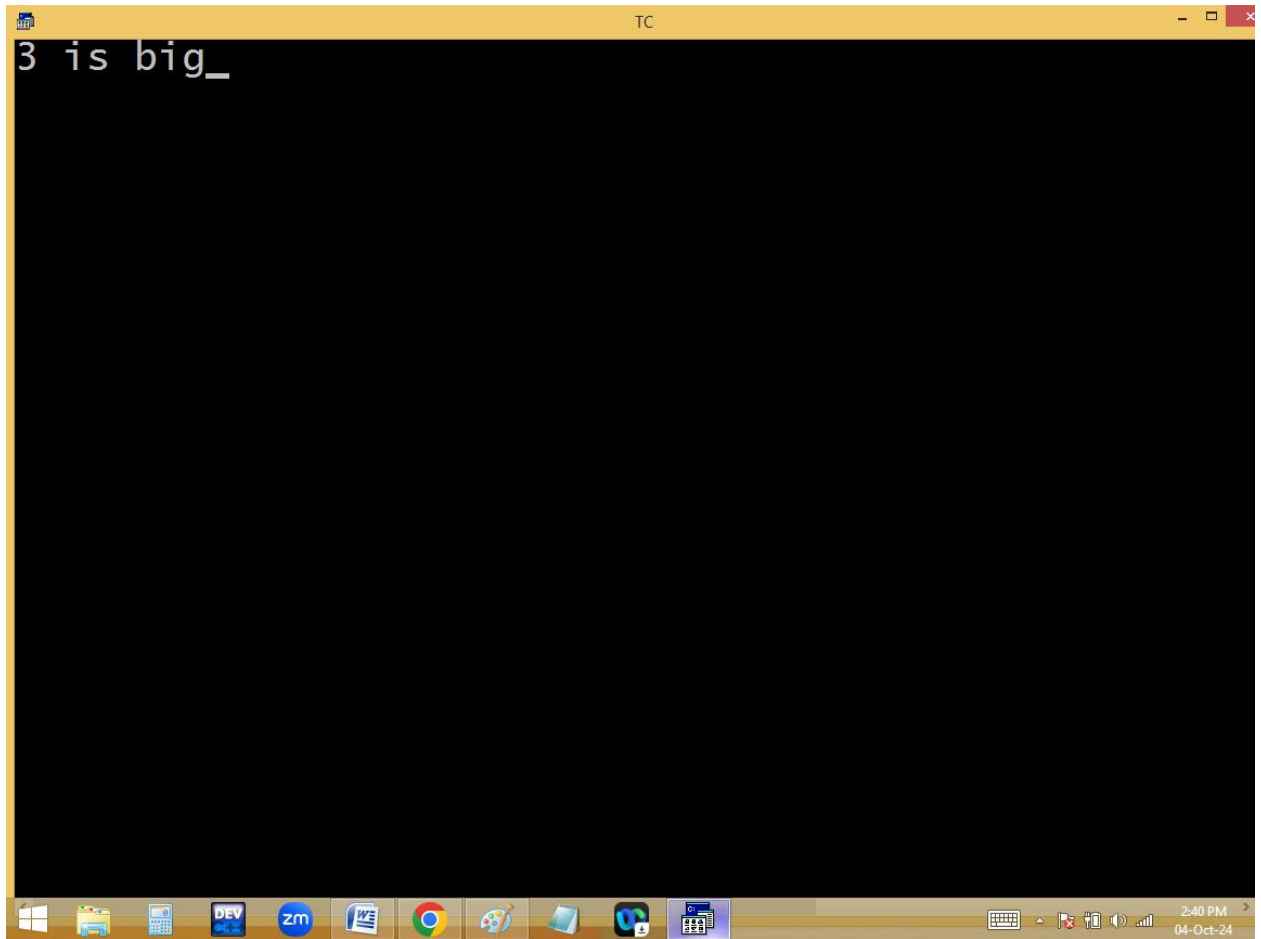


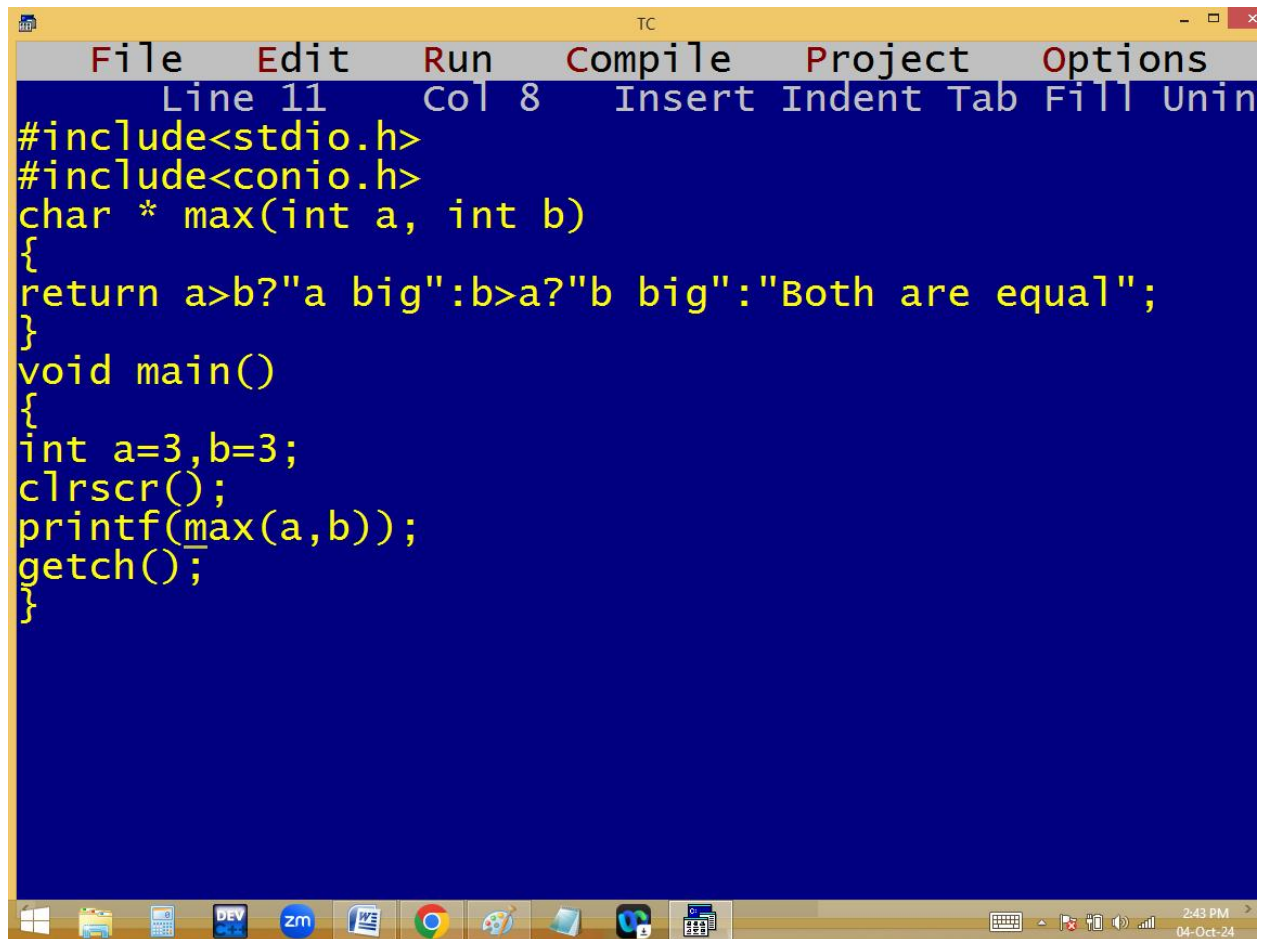


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, and Options. Under the "Run" menu, there is a submenu with the following options: Line 9, Col 8, Insert, Indent, Tab, Fill, and Unin. The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
int max(int a, int b)
{
    return a>b?a:b;
}
void main()
{
    int a=3,b=3;
    clrscr();
    printf("%d is big",max(a,b));
    getch();
}
```

At the bottom of the window is a Windows taskbar. It contains several icons: the Windows Start button, a folder icon, a calculator icon, a "DEV" icon, a "zm" icon, a "WZ" icon, a Google Chrome icon, a paint icon, a file explorer icon, a VS Code icon, and a calendar icon. On the right side of the taskbar, there is a system tray showing the time as "2:40 PM" and the date as "04-Oct-24".

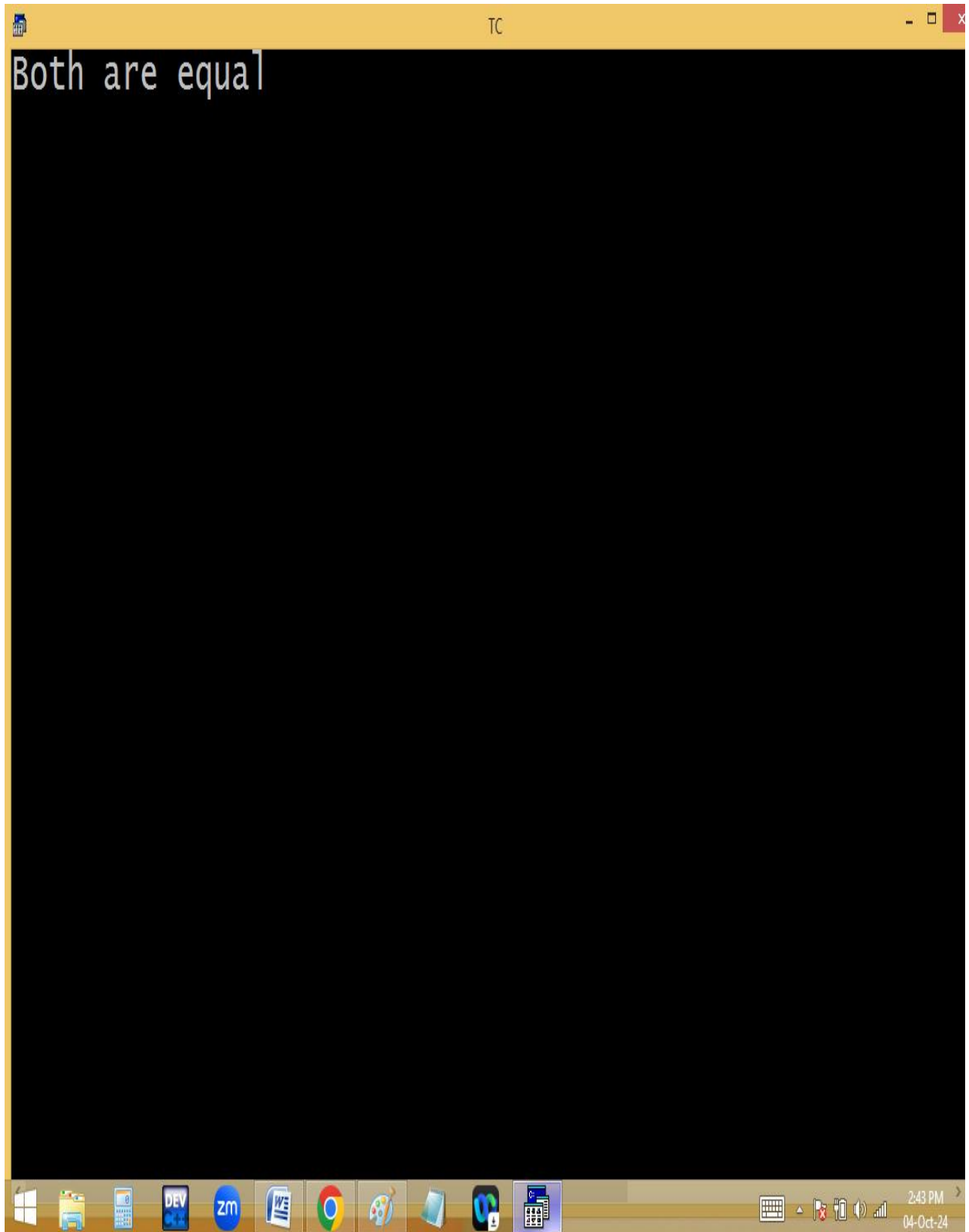




The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". Below it is a menu bar with the following options: File, Edit, Run, Compile, Project, and Options. Under the "Run" menu, there is a submenu with the following options: Line 11, Col 8, Insert, Indent, Tab, Fill, and Unin. The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
char * max(int a, int b)
{
return a>b?"a big":b>a?"b big":"Both are equal";
}
void main()
{
int a=3,b=3;
clrscr();
printf(max(a,b));
getch();
}
```

At the bottom of the window is a Windows taskbar. It contains several icons: the Start button, a folder icon, a calculator icon, a "DEV" icon, a "zm" icon, a "WZ" icon, the Google Chrome icon, a paint icon, a file explorer icon, a VS Code icon, and a calendar icon. On the right side of the taskbar, there is a system tray showing the time as 2:43 PM and the date as 04-Oct-24.



POINTERS

Pointer is a variable, which holds the address of another variable of same type.

Pointer is a memory location, which holds the address of another memory location.

Pointer is a derived data type.

Advantages:

1. Dynamic memory allocation.
2. Program performance is increased due to preventing memory wastage.
3. They are very much used in System programming.
4. They are very much used in dynamic linked list & Stacks [data structures].
5. **It allows to access local variable outside the function i.e. data sharing between functions. [call by address/Reference].**
6. **To handle strings, arrays etc in functions we need pointers.**

7. To handle **data files** we are using pointers.
8. They directly works on variable address. Due to this search time is reduced and execution speed is increased.

Dis-advantage:

They are not secured.

Syntax:

datatype * variable;

- * indicates it is a pointer data type.
- * is called indirection operator.
- * is called dereferencing operator.
- *** is a re-direction operator.**
- * indicates value at that address.
- * indicates pointer value.

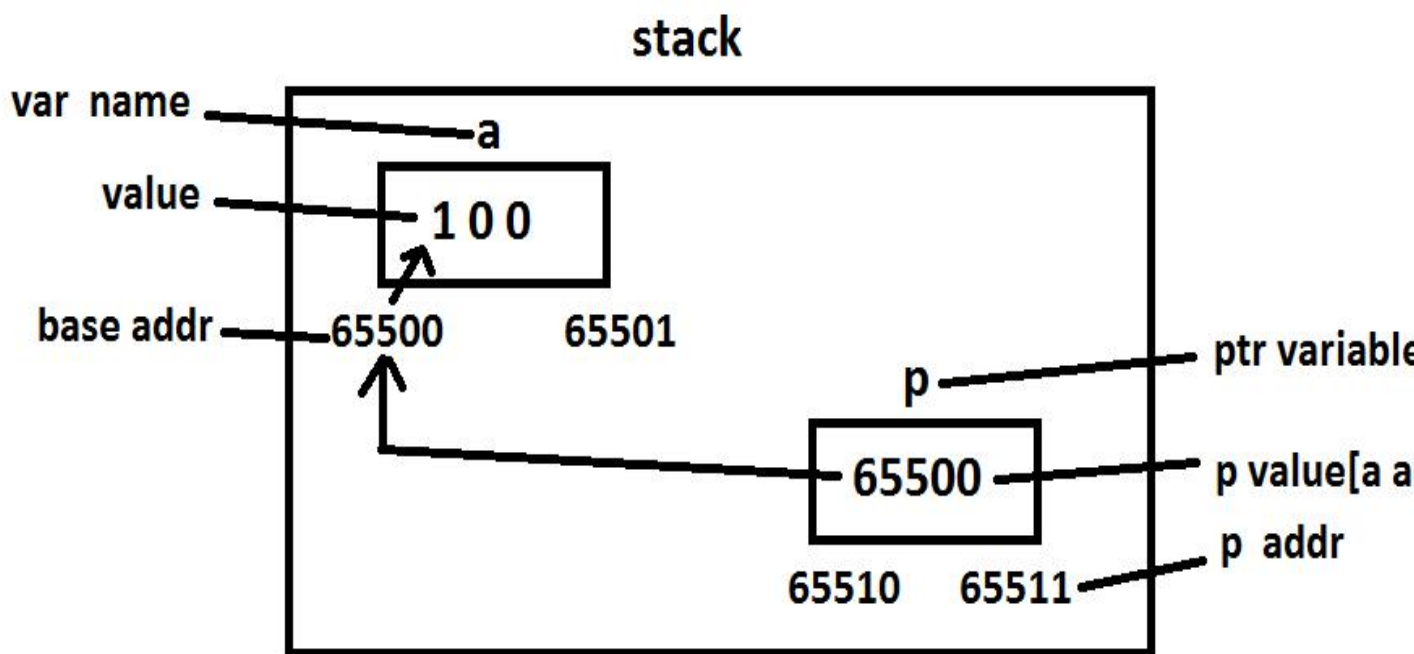
Eg:

```
int a=100, * p;
```

In the above example '**a**' is a general variable.

* indicates '**p**' is a pointer type variable and it is able to store the address of general variable '**a**' as follows.

```
p = &a;
```



In the above example, to pick the value of **a** through pointer variable **p**, we have to use the **printf()** as follows.

```
printf( "%d", *p );
```

→ **100**

Here ***p** means **value of p** or **value at that addr.**
i.e. **65500**. But **65500** is the **addr of 'a'**. The
value in **a** address is **100**.

Or

Here **p** means **65500**. ***p** means value at 65500.
i.e. **100**.

Due to this example any changes conducted in
***p** effects the value of **'a'**. Hence **p** is called
pointer to a.

Eg: ***p=200;**

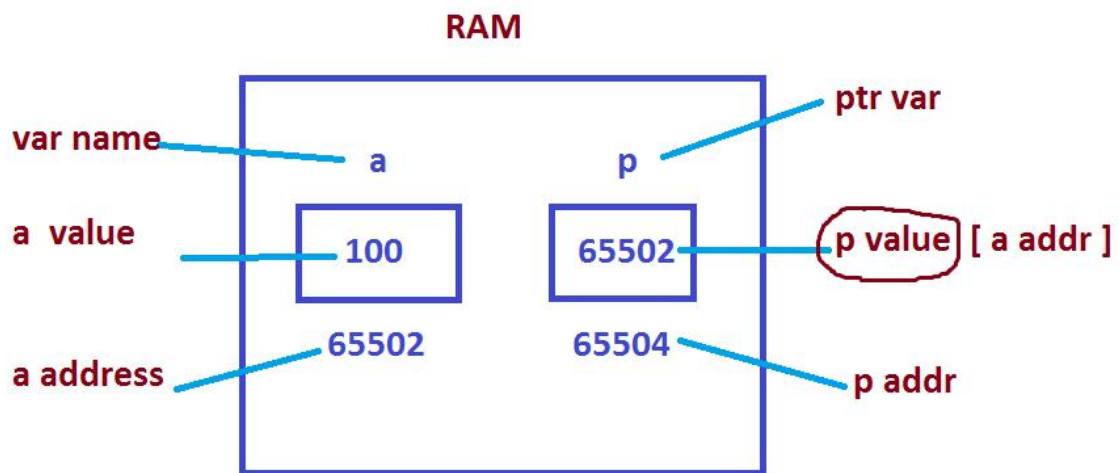
Now **a** value becomes **200**.

Eg:

**Finding a variable value and address using a
pointer:**

```
TC
File Edit Run Compile Project Options
Line 14 Col 1 Insert Indent Tab Fill Unin
#include<stdio.h>
#include<conio.h>
void main()
{
int a=100,*p; /* ptr declaration */
p=&a; /* initialization*/
clrscr();
printf("a value %d\n",a);
printf("a addr %u\n",&a);
printf("p value %u\n",p);
printf("a value %d\n",*p);
*p = 200;
printf("a=%d, *p=%d\n",a,*p);
a = 300;
printf("a=%d, *p=%d\n",a,*p);
getch();
}
```

```
TC
a value 100
a addr 65502
p value 65502
a value 100
a=200, *p=200
a=300, *p=300
```

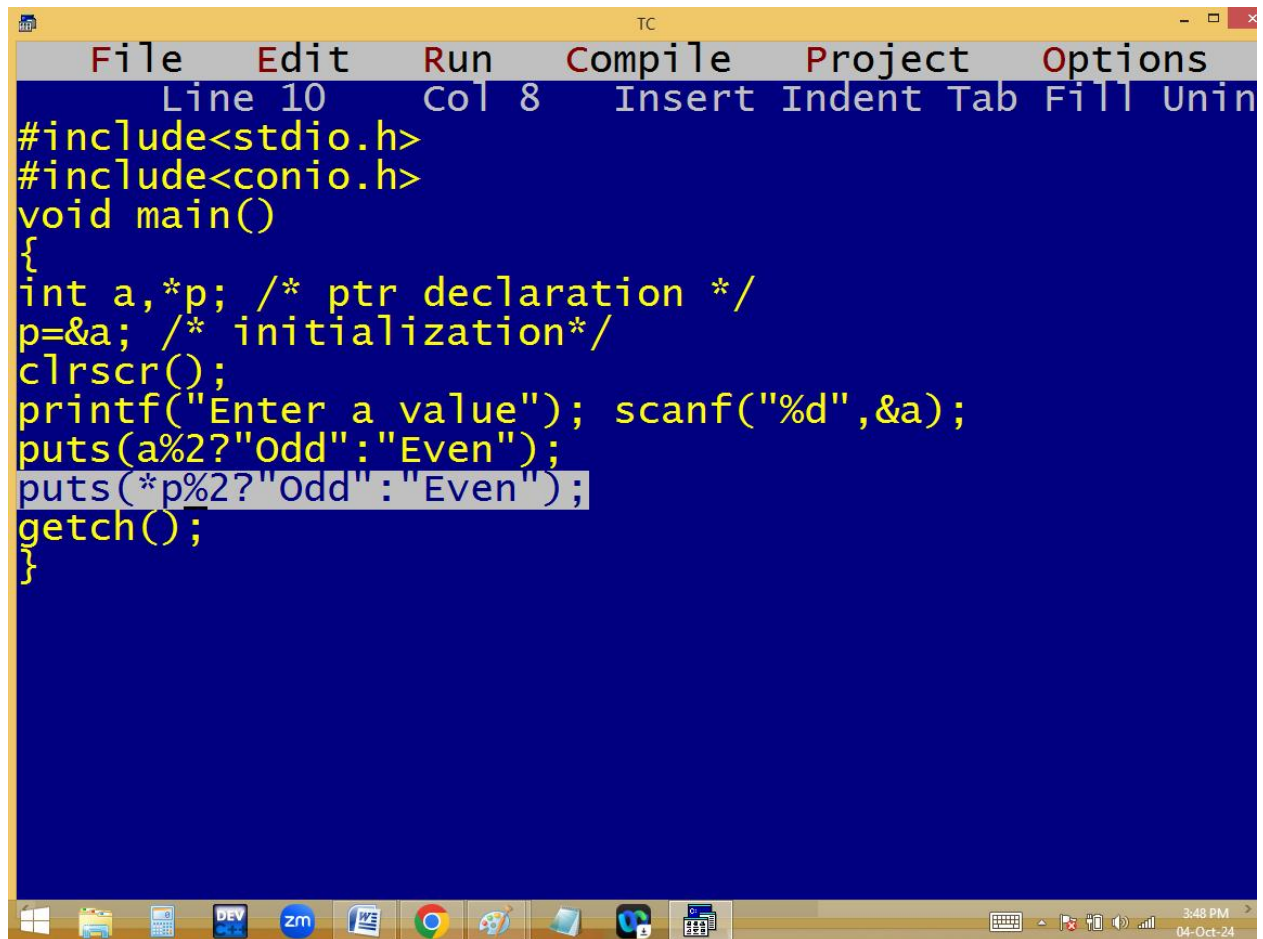


```
printf("%d", * p );
```

↳ 65502

↳ * 65502 ==> value at 65502 ==> 100

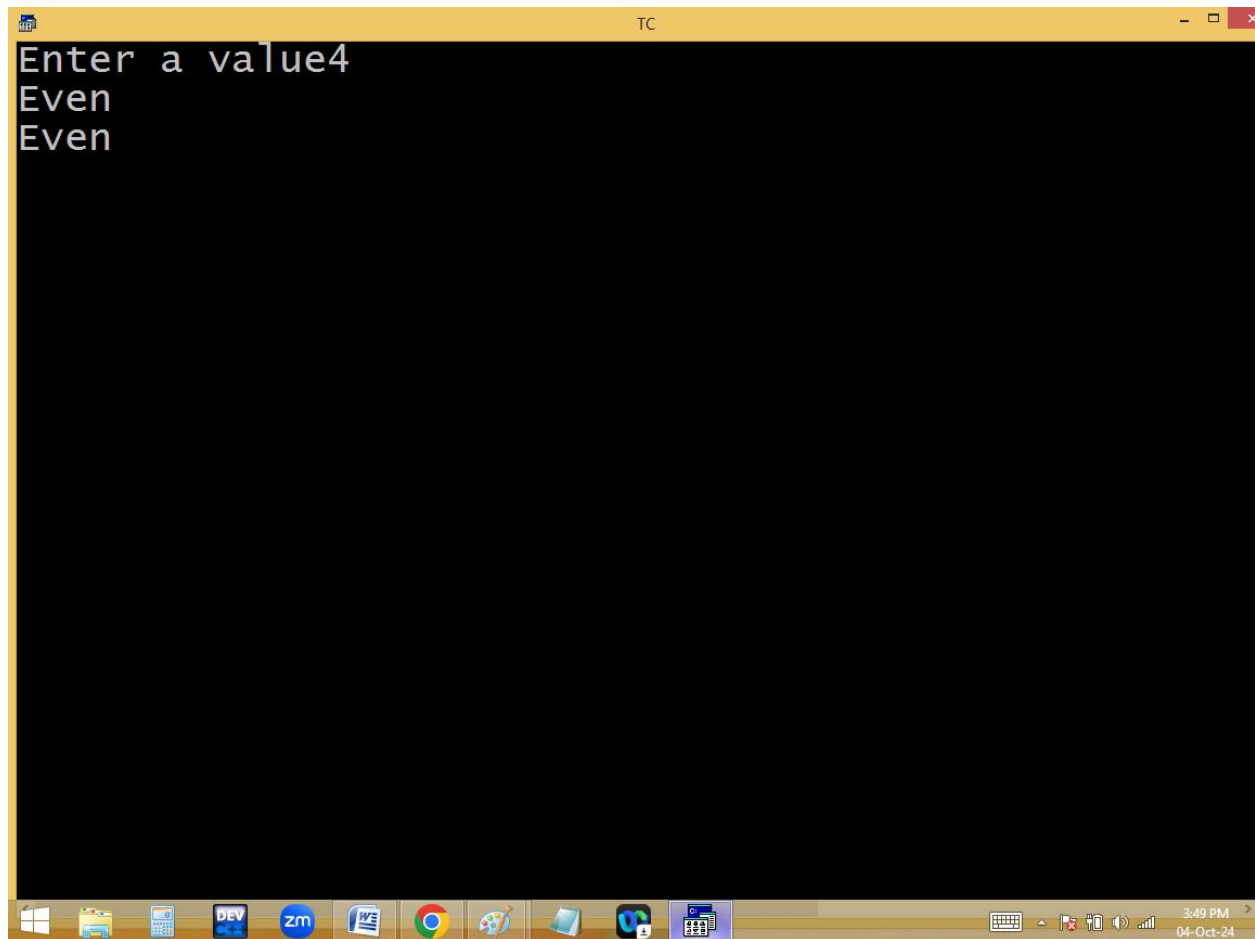
Finding even/odd using pointer:



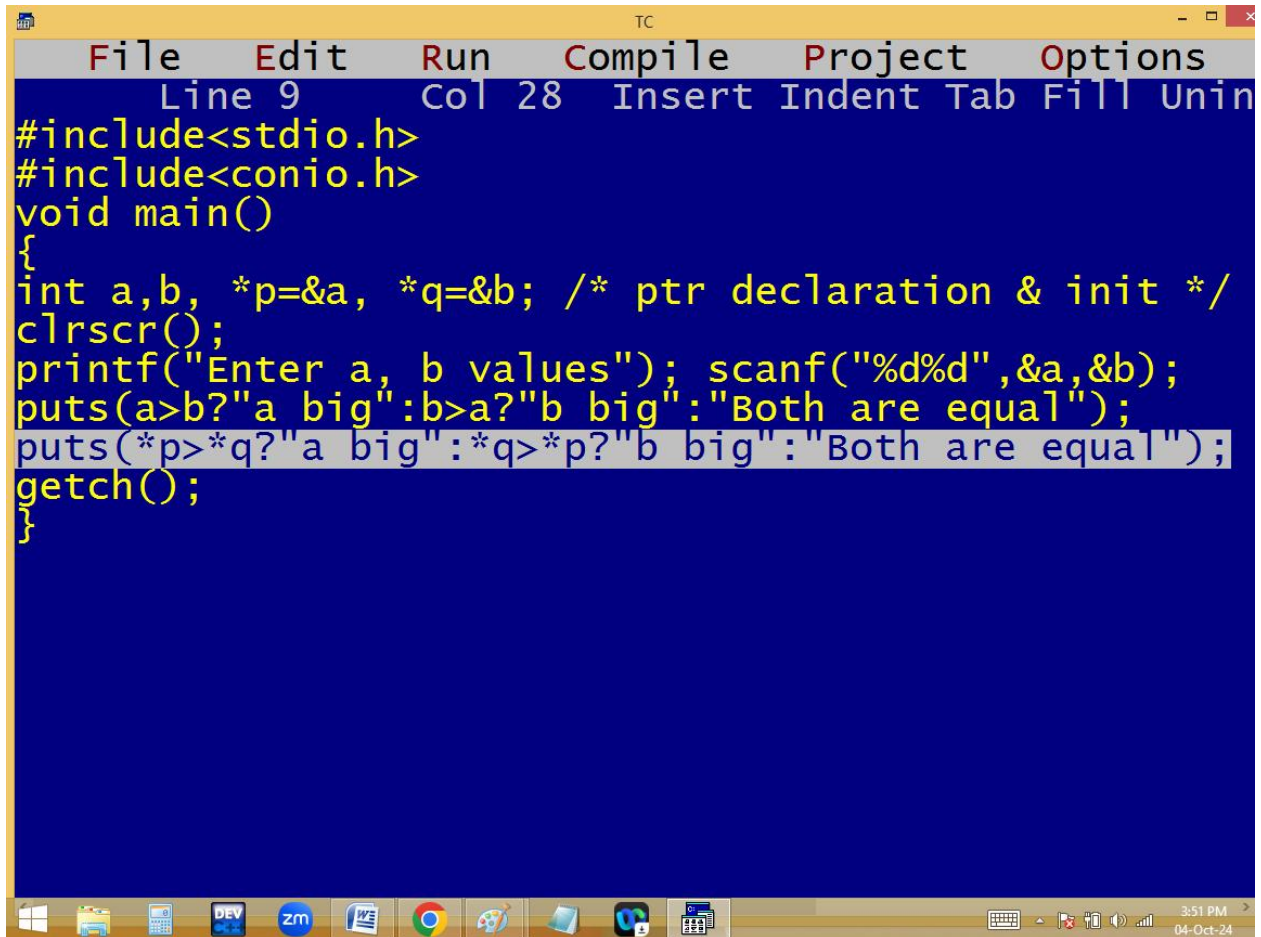
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", and "Options". The status bar at the top indicates "Line 10", "Col 8", and "Insert Indent Tab Fill Unin". The code is written in yellow text on a dark blue background. It includes headers for `stdio.h` and `conio.h`, and defines a `main` function. Inside `main`, it declares an integer `a` and a pointer `p`, initializes `p` to point to `a`, clears the screen with `clrscr()`, prompts the user to enter a value, reads the value into `a` using `scanf`, and then prints the parity of both `a` and `*p` using `puts`. The line `puts(*p%2?"Odd":"Even");` is highlighted with a light blue selection. The code ends with `getch()` and a closing brace for `main`. The Windows taskbar at the bottom shows various icons including the Start button, File Explorer, Calculator, DEV C++, Zoom, Word, Chrome, and others. The system clock in the bottom right corner shows "3:48 PM" and "04-Oct-24".

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a,*p; /* ptr declaration */
p=&a; /* initialization*/
clrscr();
printf("Enter a value"); scanf("%d",&a);
puts(a%2?"Odd":"Even");
puts(*p%2?"Odd":"Even");
getch();
}
```

```
TC
Enter a value4
Even
Even
```



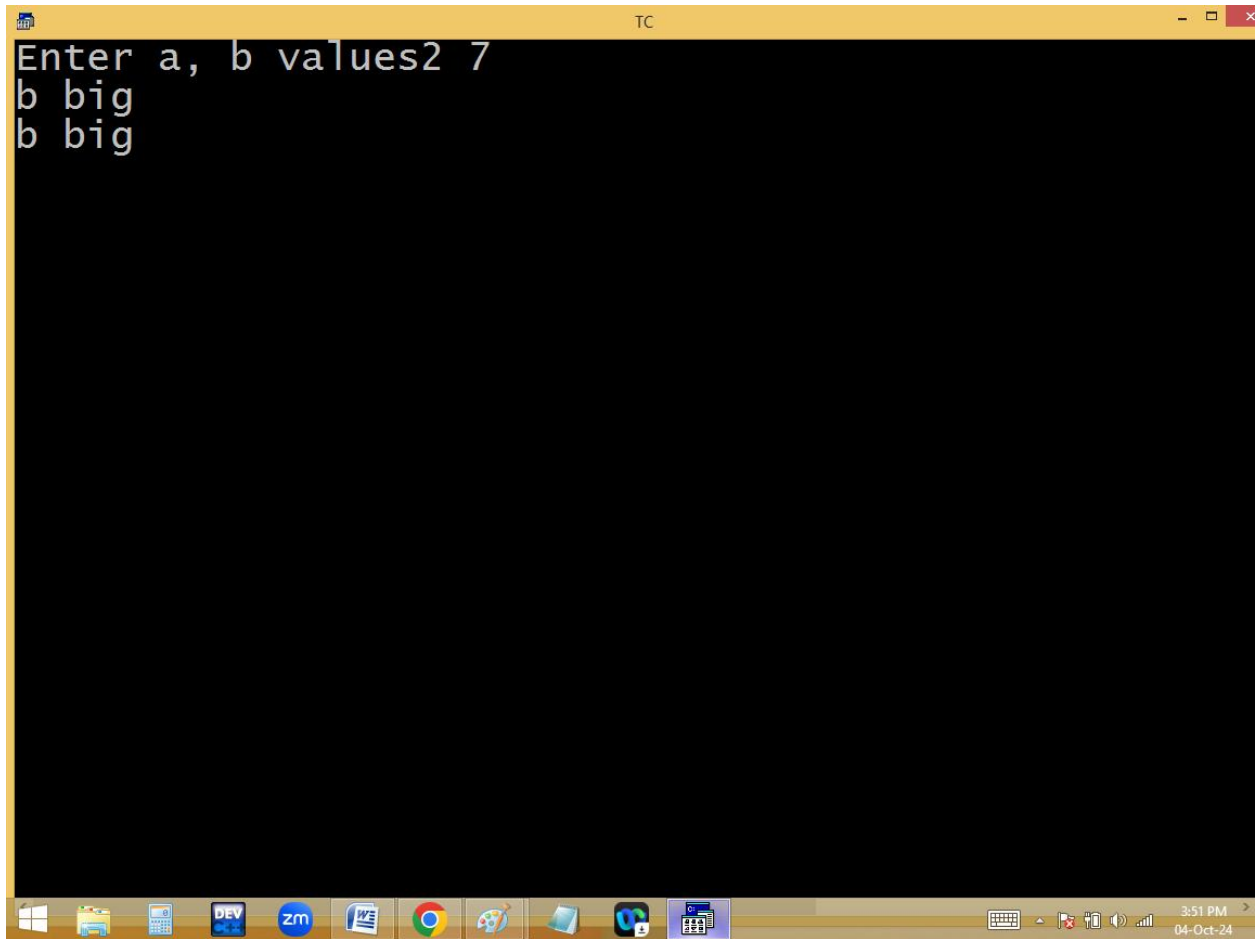
Finding max in 2 no's using pointers:

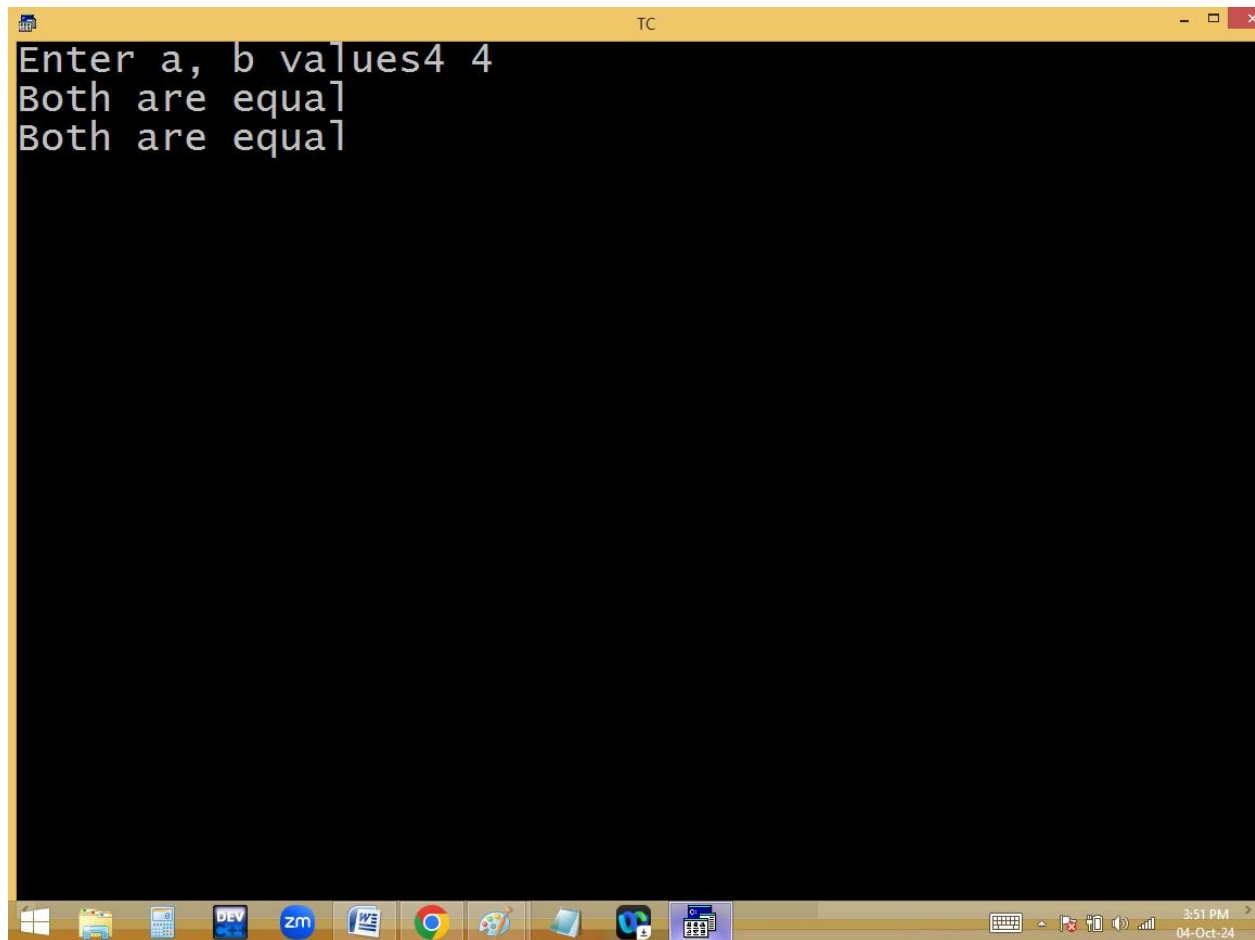


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, and Options. Under the "Run" menu, there is a submenu with the following options: Line 9, Col 28, Insert, Indent, Tab, Fill, and Unin. The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b, *p=&a, *q=&b; /* ptr declaration & init */
clrscr();
printf("Enter a, b values"); scanf("%d%d",&a,&b);
puts(a>b?"a big":b>a?"b big":"Both are equal");
puts(*p>*q?"a big":*q>*p?"b big":"Both are equal");
getch();
}
```

The bottom of the window shows a Windows taskbar with various icons, including the Start button, File Explorer, Calculator, DEV C++, Zoom, Word, Chrome, and others. The system clock in the bottom right corner shows the time as 3:51 PM and the date as 04-Oct-24.



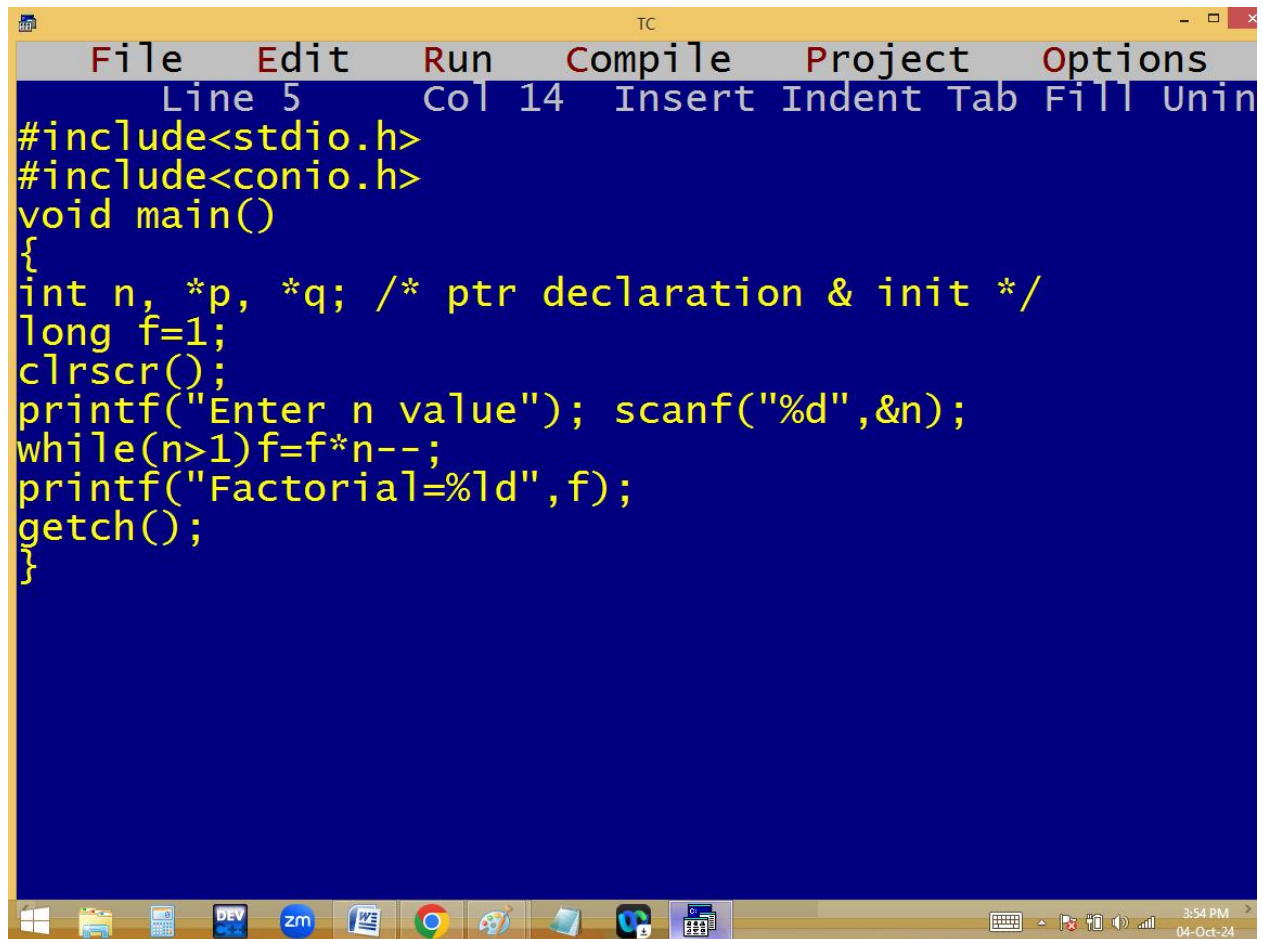


The image shows a screenshot of a Turbo C++ (TC) compiler window. The window has a yellow title bar with the text "TC" in the center. The main area is black with white text. The text displayed is:

```
Enter a, b values4 4  
Both are equal  
Both are equal
```

The window is running on a Windows operating system, as evidenced by the taskbar at the bottom. The taskbar includes icons for various applications: Windows Start button, File Explorer, Calculator, DEV C++, ZOOM, Word, Chrome, Paint, a folder icon, a game icon, and a calculator icon. The system tray on the right shows the time as 3:51 PM and the date as 04-Oct-24.

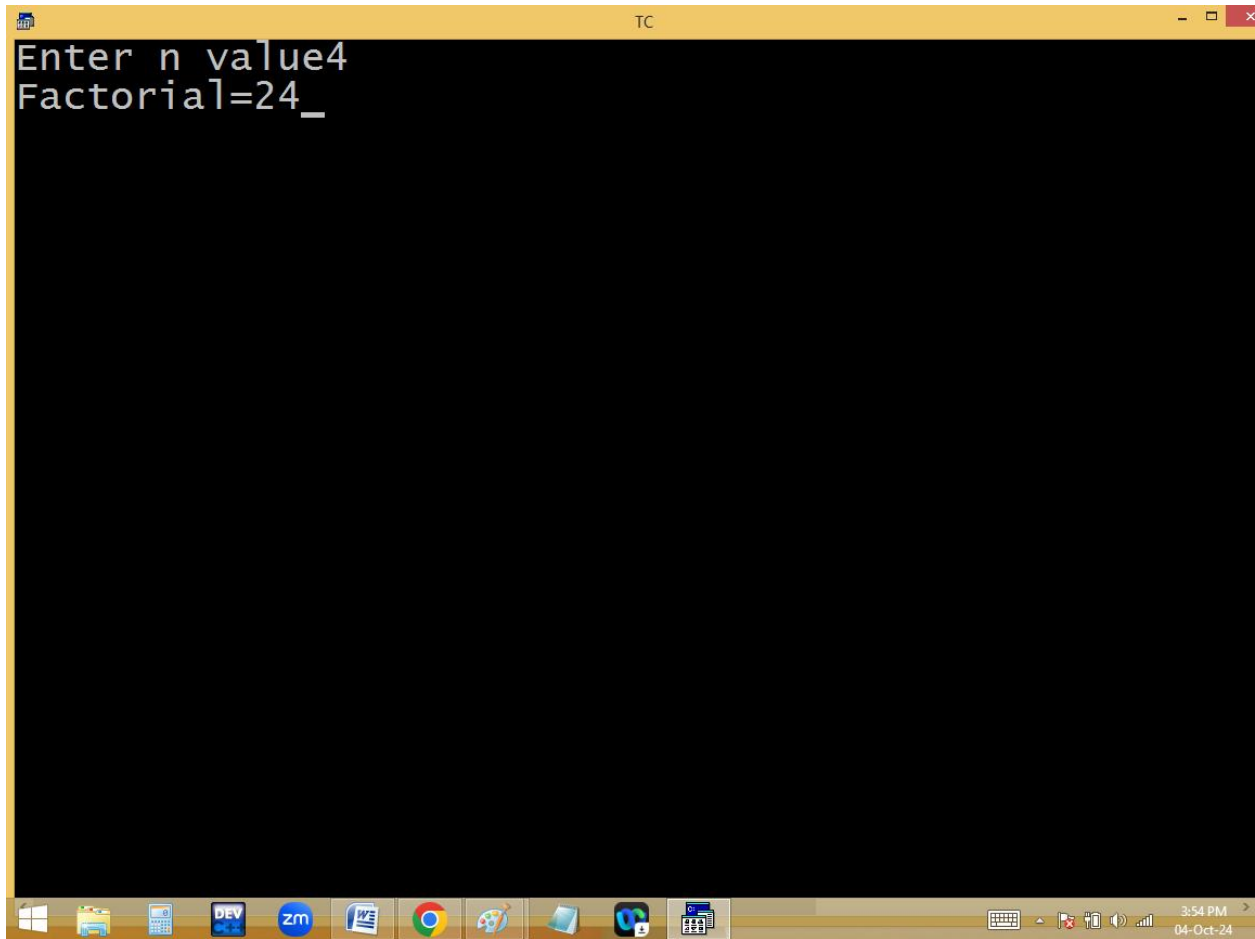
Finding factorial using pointer:

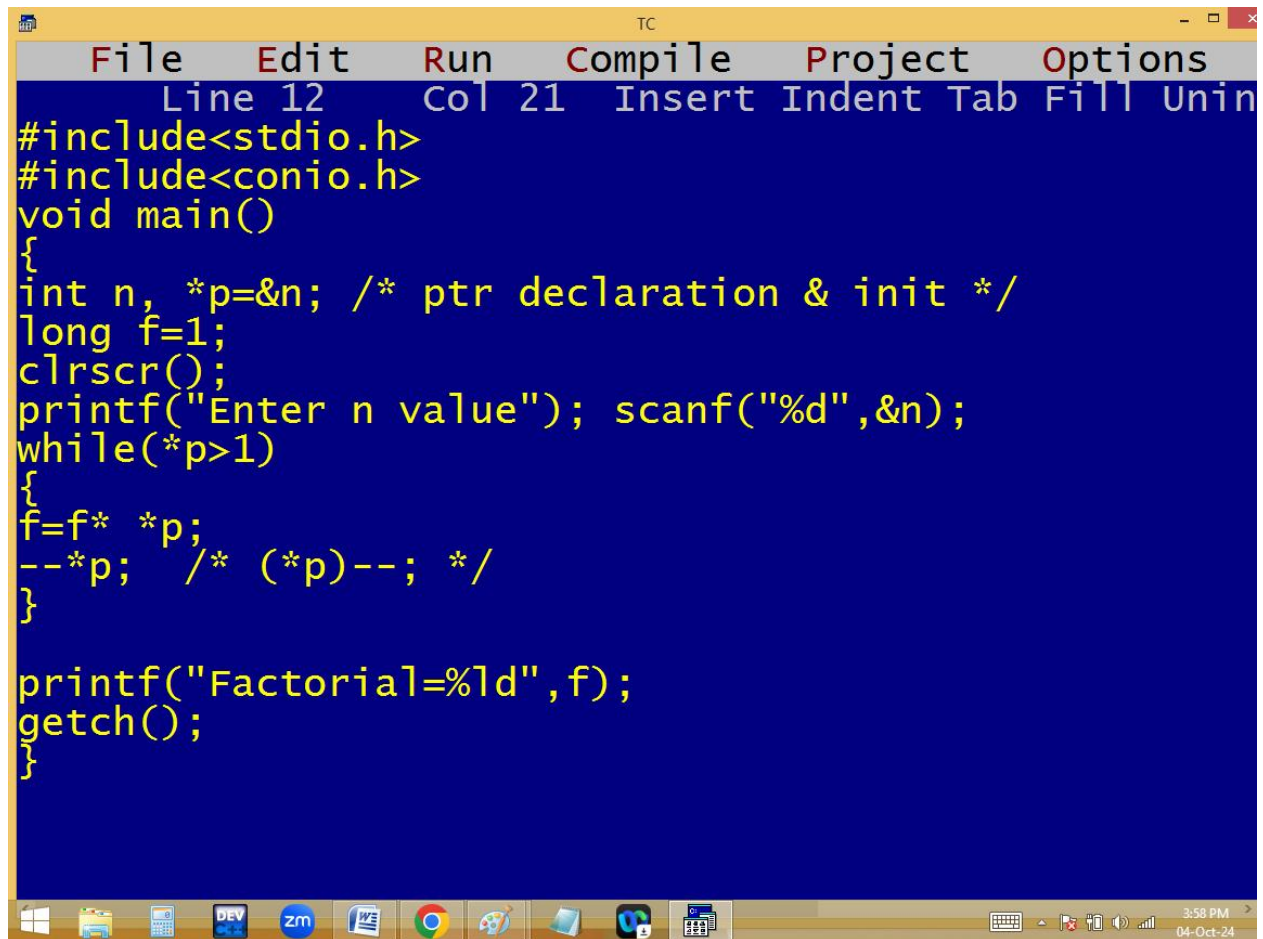


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". Below it is a menu bar with the following options: File, Edit, Run, Compile, Project, and Options. A status bar at the top of the editor area displays "Line 5", "Col 14", and "Insert Indent Tab Fill Unin". The main editor area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n, *p, *q; /* ptr declaration & init */
long f=1;
clrscr();
printf("Enter n value"); scanf("%d",&n);
while(n>1)f=f*n--;
printf("Factorial=%ld",f);
getch();
}
```

At the bottom of the screen is a Windows taskbar. It contains several icons: the Start button, File Explorer, Calculator, DEV C++, Zoom (zm), Word (W), Google Chrome, a game controller icon, a folder icon, a Teams icon, and a calendar icon. On the right side of the taskbar, the system clock shows "3:54 PM" and the date "04-Oct-24".





The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, and Options. Under the "Run" menu, there is a submenu with the following options: Line 12, Col 21, Insert, Indent, Tab, Fill, and Unin. The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n, *p=&n; /* ptr declaration & init */
long f=1;
clrscr();
printf("Enter n value"); scanf("%d",&n);
while(*p>1)
{
f=f* *p;
--*p; /* (*p)--; */
}

printf("Factorial=%ld",f);
getch();
}
```

At the bottom of the window is a Windows taskbar. It contains several icons: the Start button, a folder icon, a calculator icon, a "DEV" icon, a "zm" icon, a "WZ" icon, a Google Chrome icon, a Paint icon, a file explorer icon, a folder icon, a "C" icon, and a calendar icon. On the right side of the taskbar, there is a system tray showing the time as 3:58 PM and the date as 04-Oct-24.

